

Measurement of π^0 and η mesons in oxygen-oxygen collisions at
 $\sqrt{s_{\text{NN}}} = 5.36$ TeV using Machine Learning-based photon
identification in ALICE Run 3

Supplemental figures

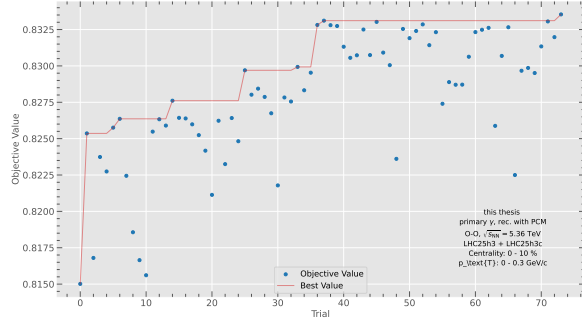
Machine-learning hyperparameter optimisation in proton-proton collisions
at $\sqrt{s} = 5.36$ TeV

by Charlize Gohmann
ML models by Dr. Ana Marin

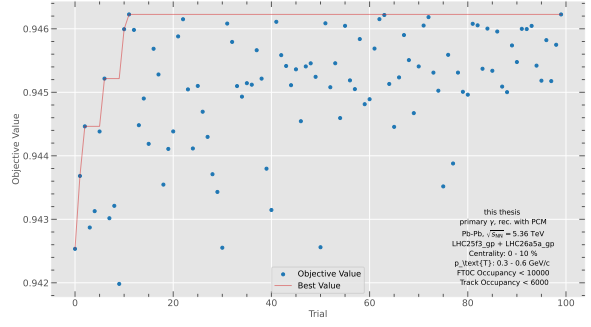
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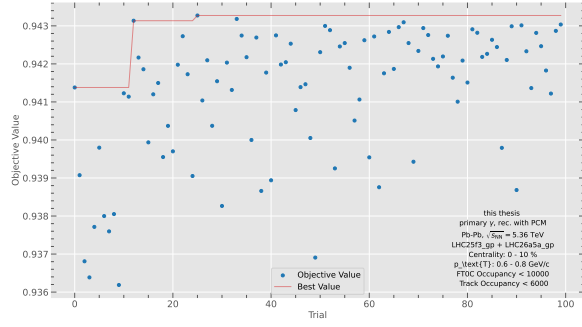
1 Hyperparameter optimisation history



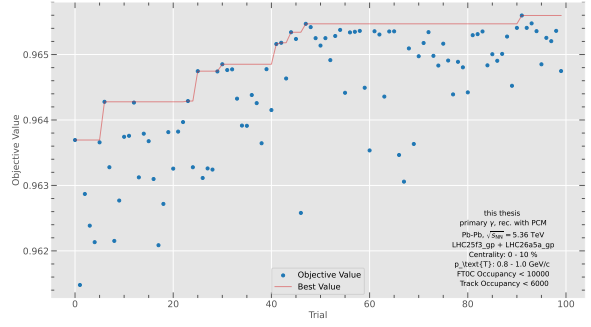
(a) 0-0.3 GeV/c



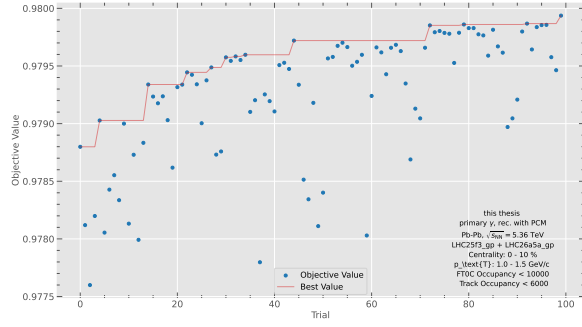
(b) 0.3-0.6 GeV/c



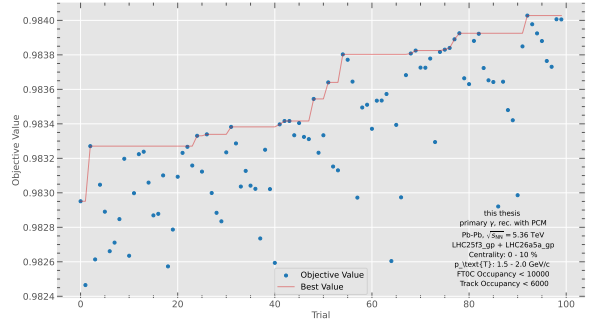
(c) 0.6-0.8 GeV/c



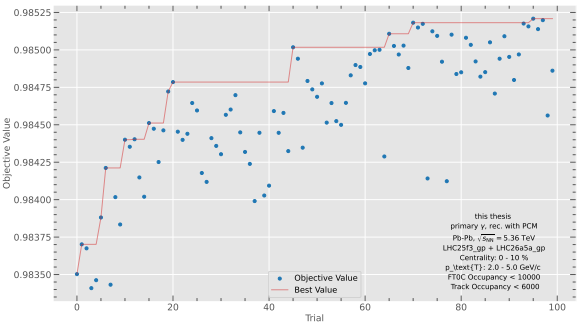
(d) 0.8-1.0 GeV/c



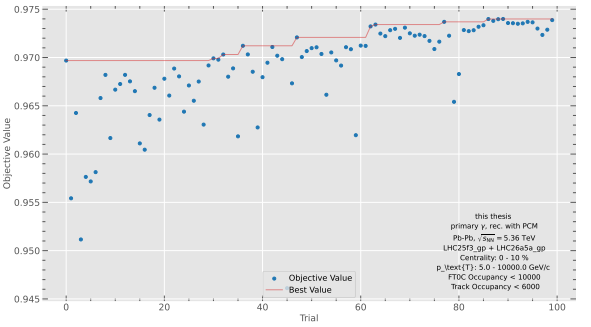
(e) 1.0-1.5 GeV/c



(f) 1.5-2.0 GeV/c

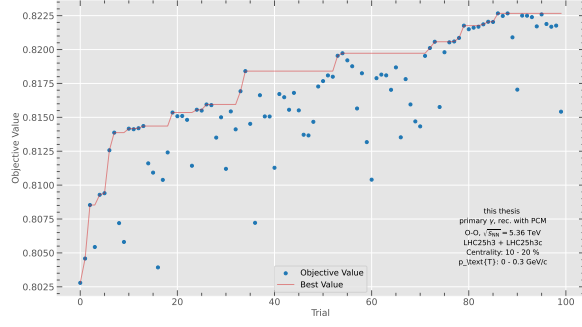


(g) 2.0-5.0 GeV/c

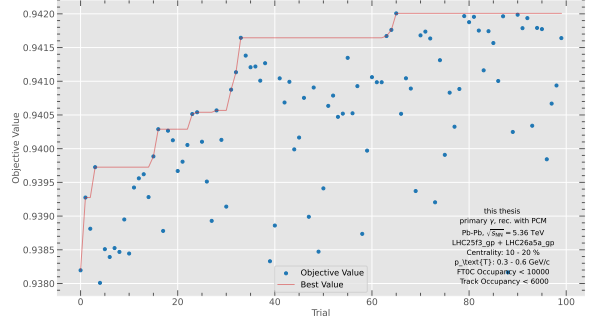


(h) 5.0-10000.0 GeV/c

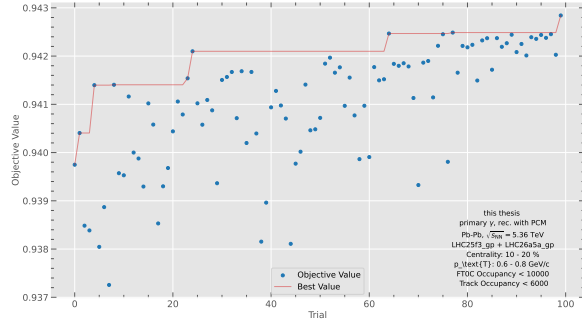
Figure 1: Hyperparameter optimisation history for all p_T bins in the centrality range 0-10 %



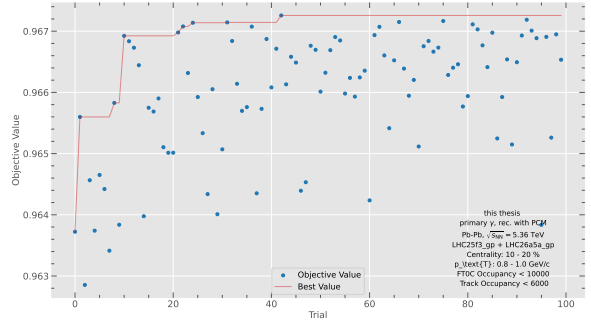
(a) 0-0.3 GeV/c



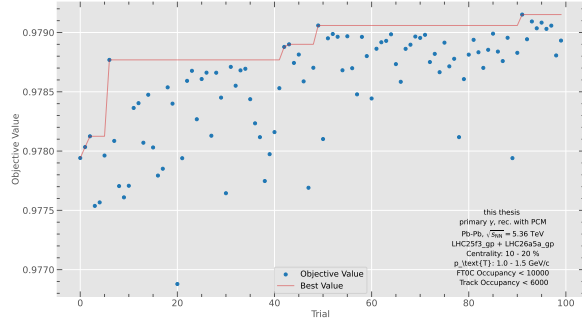
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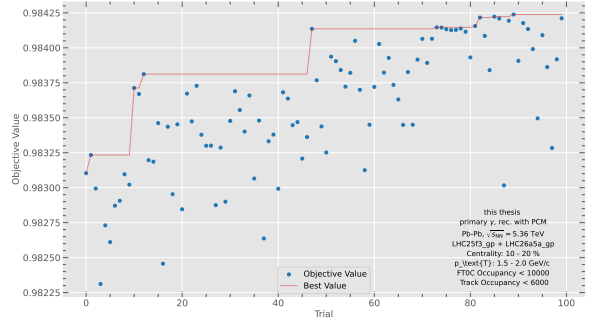
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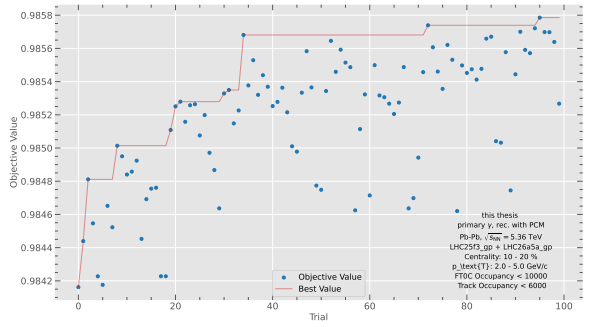
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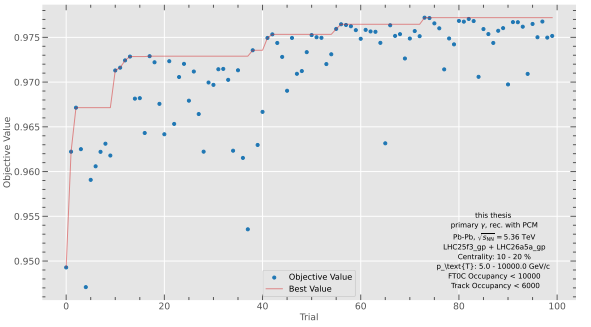
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(f) 1.5-2.0 GeV/c

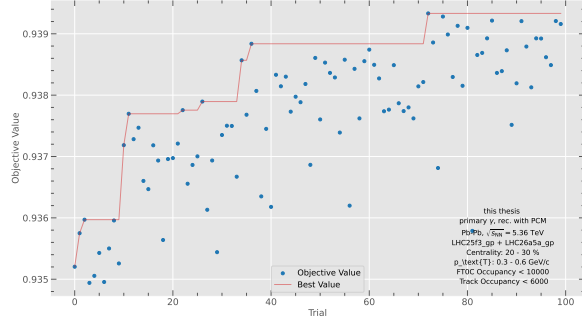


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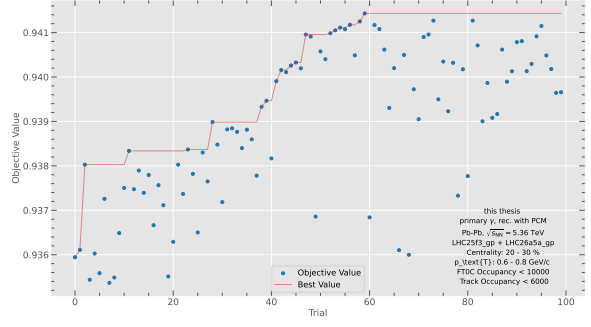


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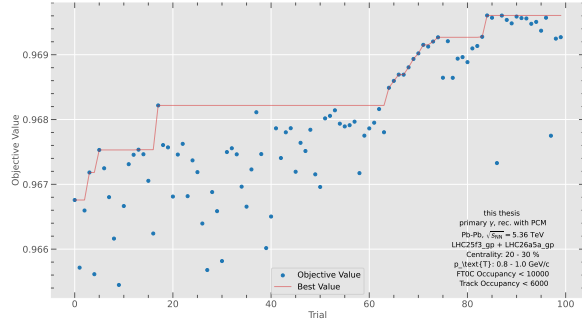
Figure 2: Hyperparameter optimisation history for all p_T bins in the centrality range 10-20 %



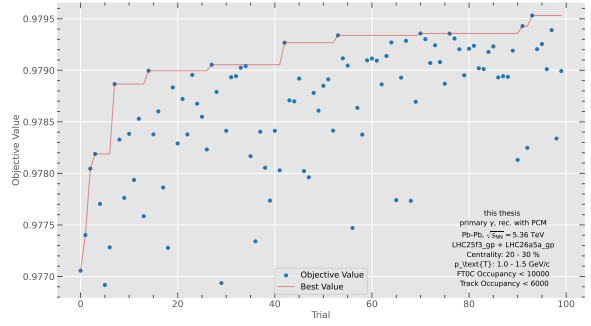
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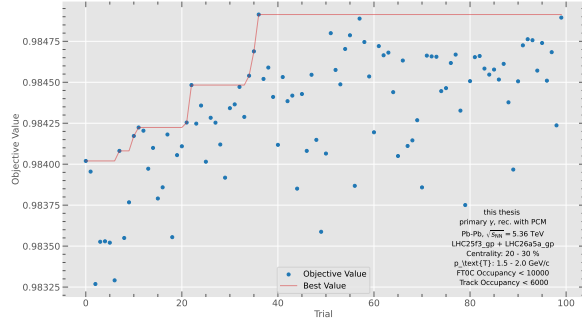
(b) 0.6-0.8 GeV/c



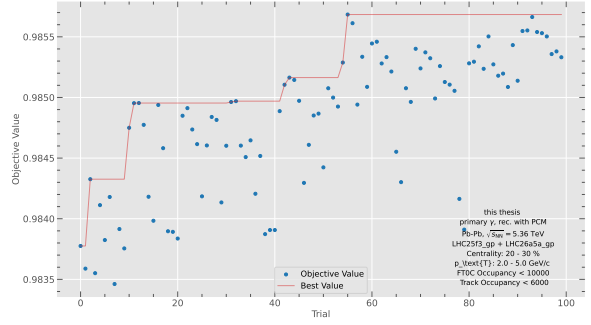
(c) 0.8-1.0 GeV/c



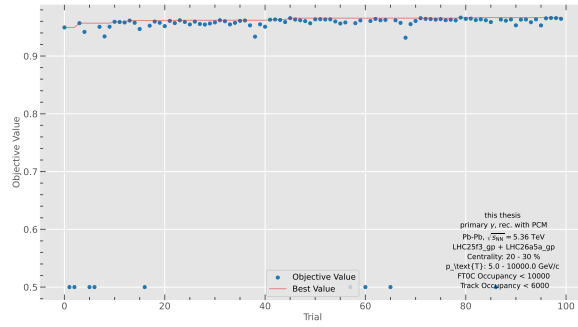
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c



(f) 2.0-5.0 GeV/c



(g) 5.0-10000.0 GeV/c

Figure 3: Hyperparameter optimisation history for all p_T bins in the centrality range 20-30 %

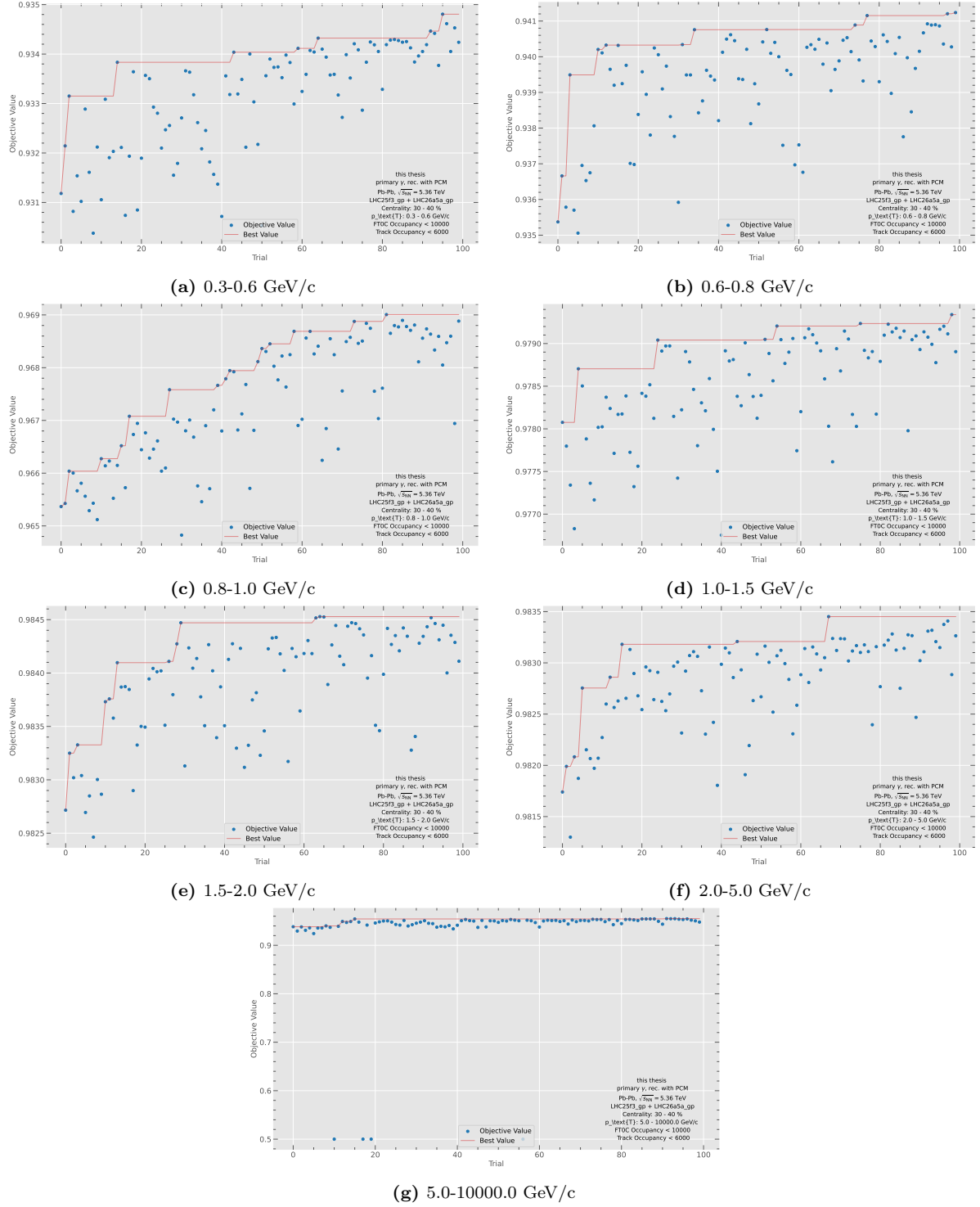
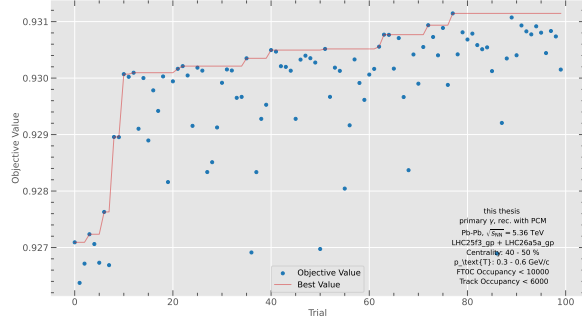
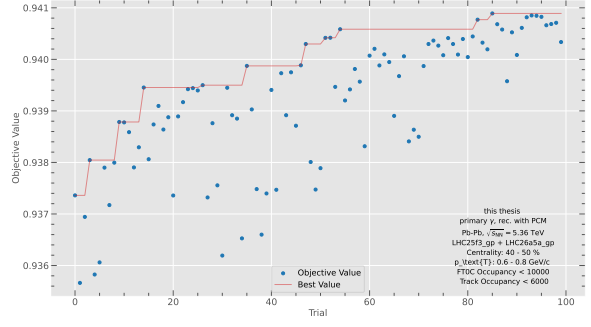


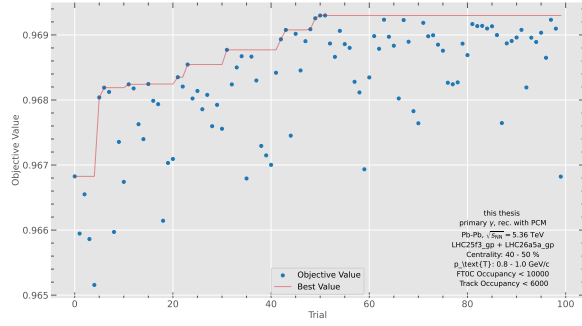
Figure 4: Hyperparameter optimisation history for all p_T bins in the centrality range 30-40 %



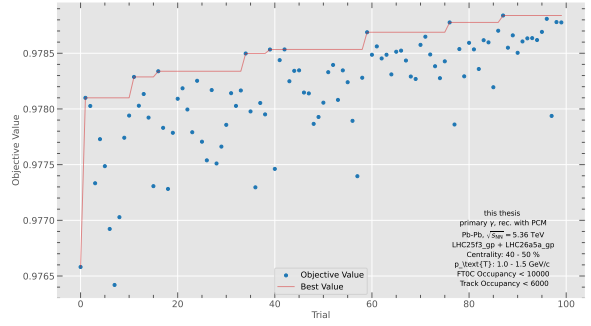
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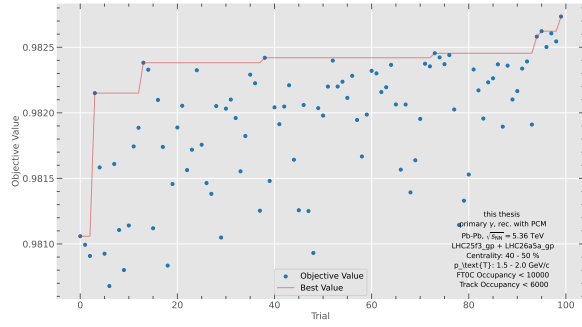
(b) 0.6-0.8 GeV/c



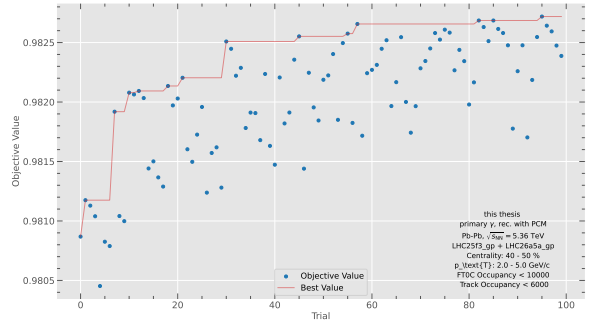
(c) 0.8-1.0 GeV/c



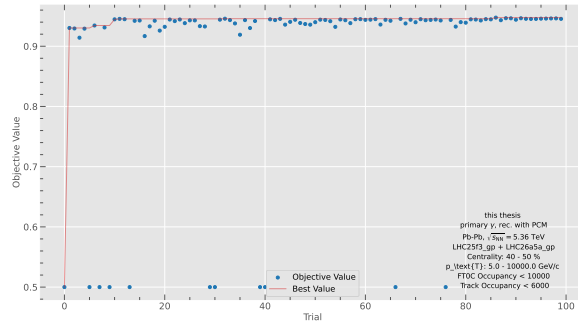
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

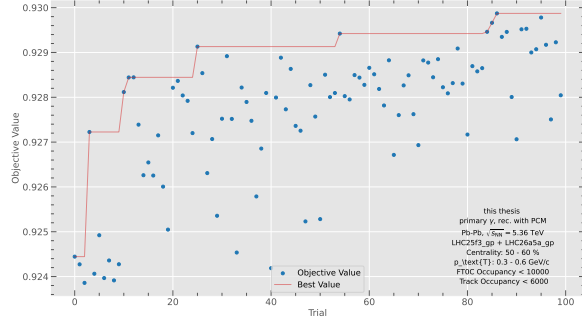


(f) 2.0-5.0 GeV/c

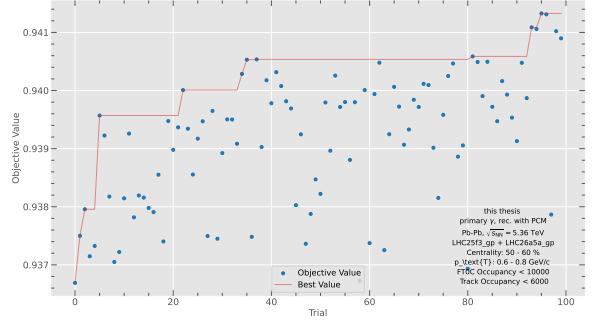


(g) 5.0-10000.0 GeV/c

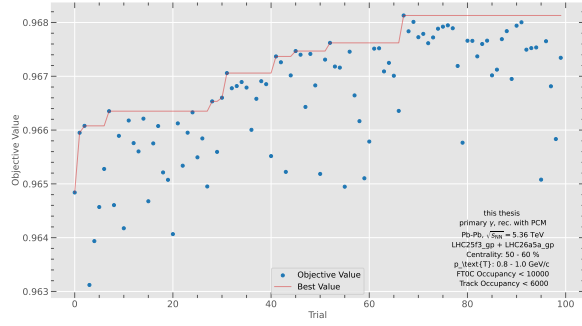
Figure 5: Hyperparameter optimisation history for all p_T bins in the centrality range 40-50 %



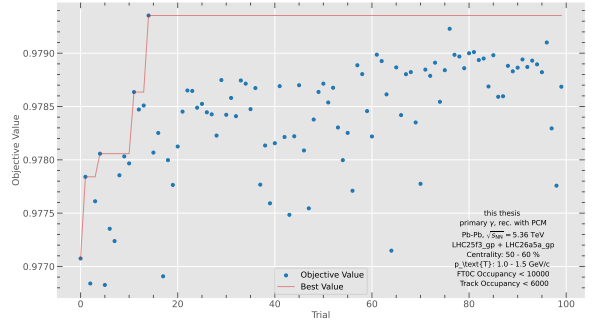
(a) 0.3-0.6 GeV/c



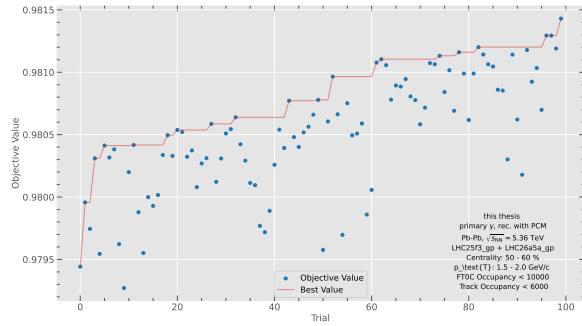
(b) 0.6-0.8 GeV/c



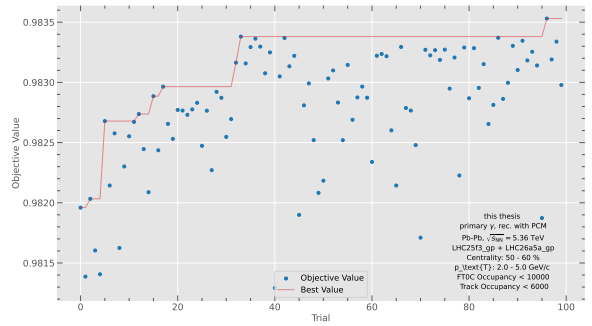
(c) 0.8-1.0 GeV/c



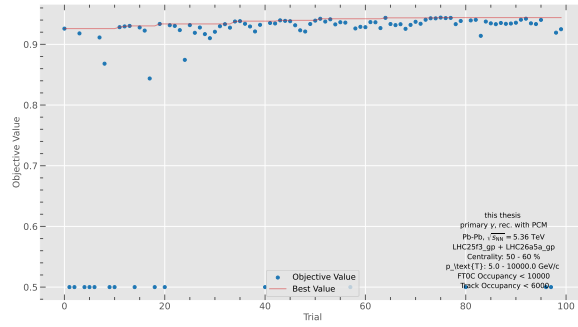
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

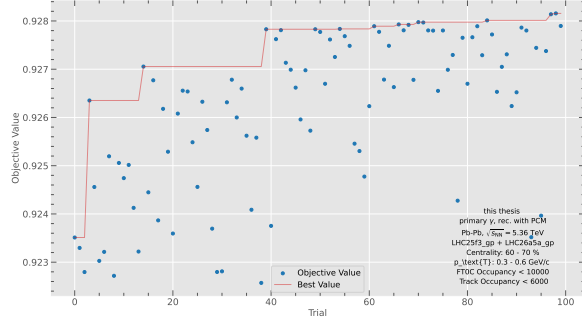


(f) 2.0-5.0 GeV/c

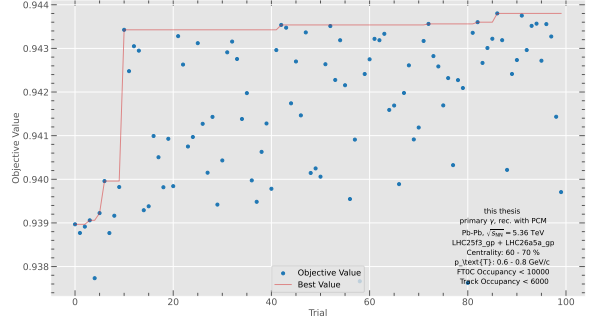


(g) 5.0-10000.0 GeV/c

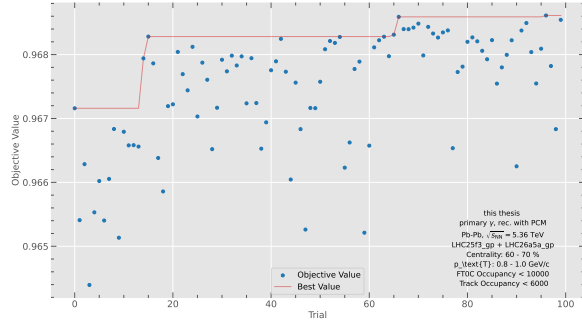
Figure 6: Hyperparameter optimisation history for all p_T bins in the centrality range 50-60 %



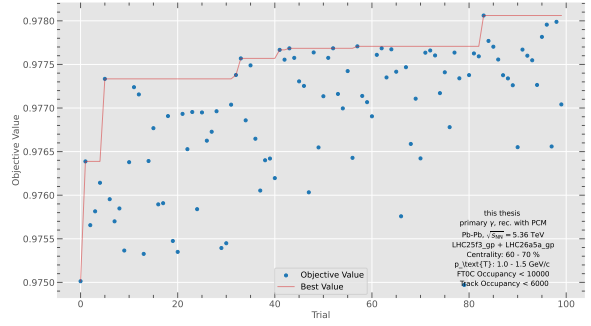
(a) 0.3-0.6 GeV/c



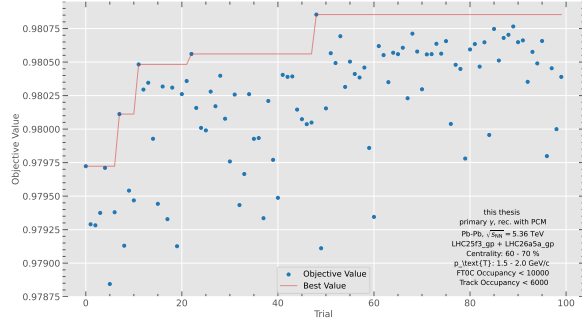
(b) 0.6-0.8 GeV/c



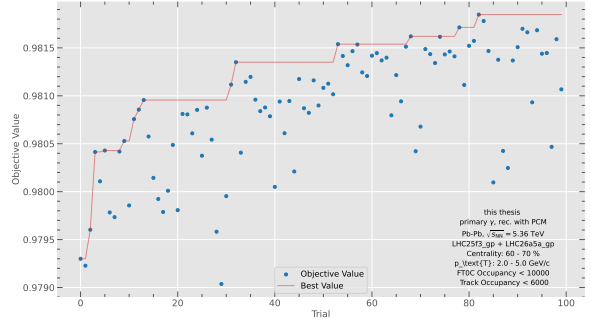
(c) 0.8-1.0 GeV/c



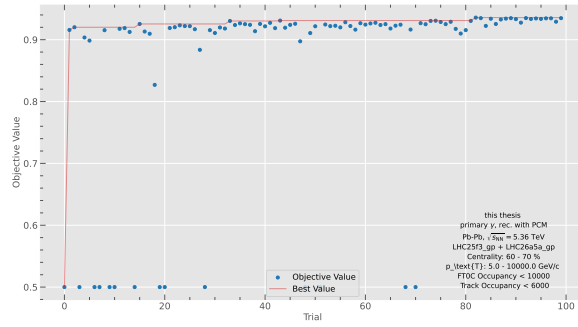
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

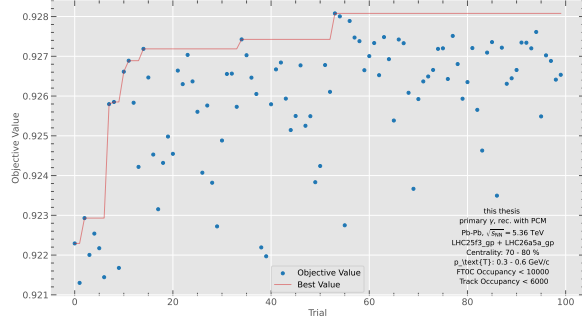


(f) 2.0-5.0 GeV/c

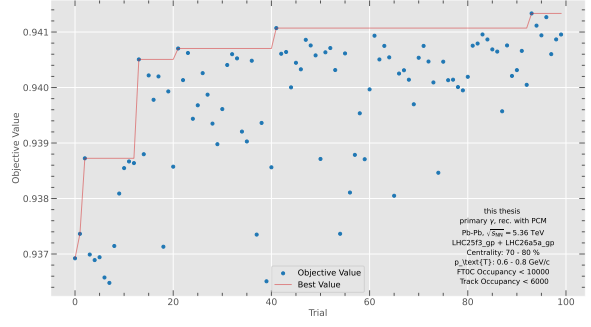


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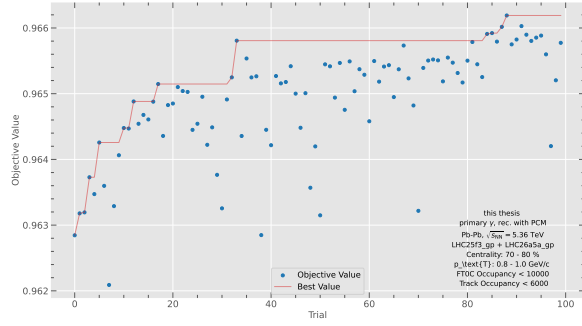
Figure 7: Hyperparameter optimisation history for all p_T bins in the centrality range 60-70 %



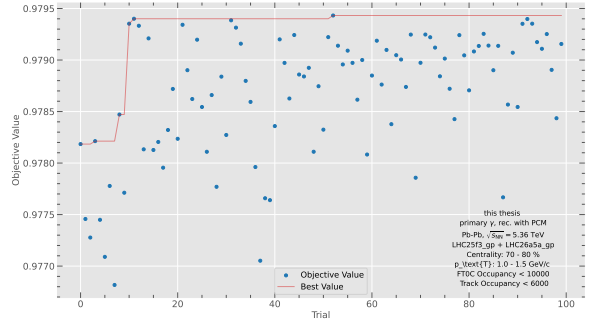
(a) 0.3-0.6 GeV/c



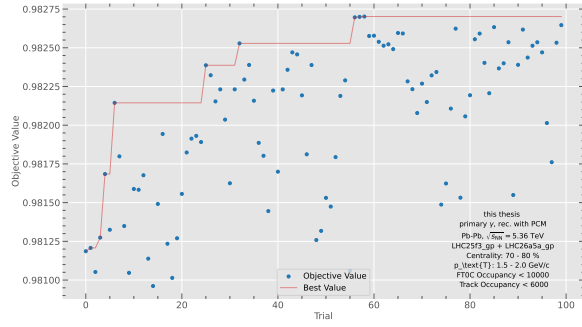
(b) 0.6-0.8 GeV/c



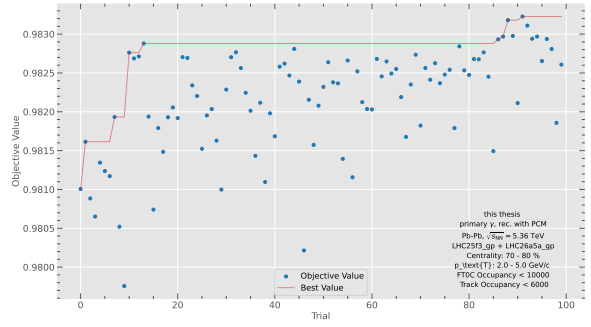
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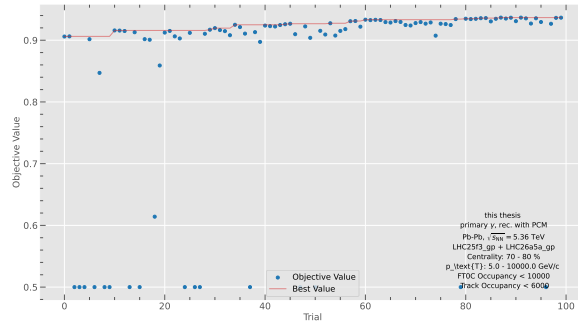
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

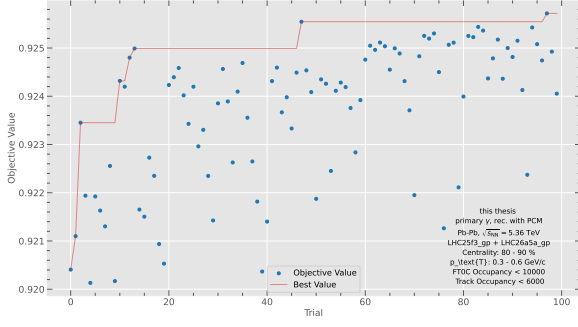


(f) 2.0-5.0 GeV/c

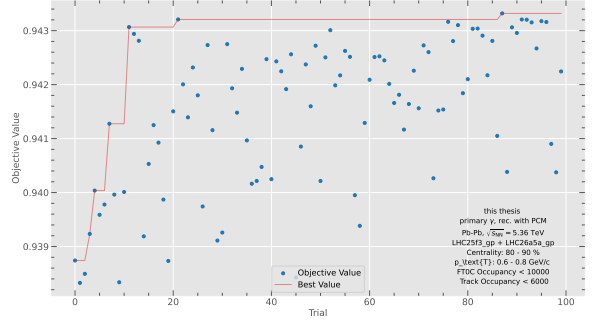


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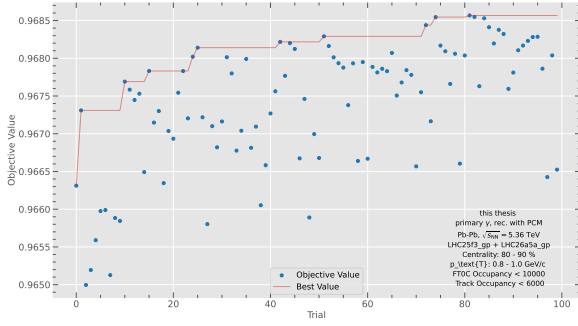
Figure 8: Hyperparameter optimisation history for all p_T bins in the centrality range 70-80 %



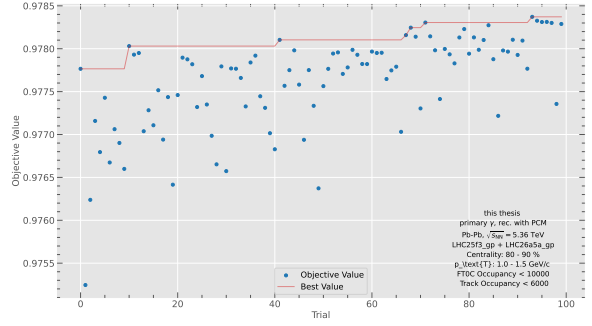
(a) 0.3-0.6 GeV/c



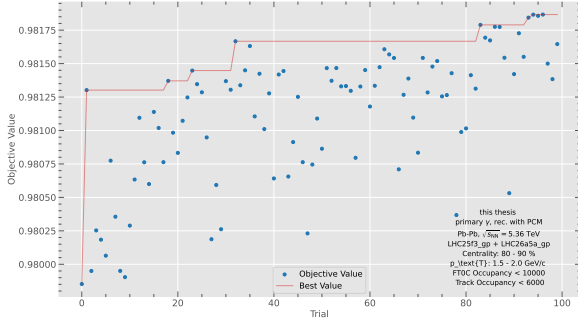
(b) 0.6-0.8 GeV/c



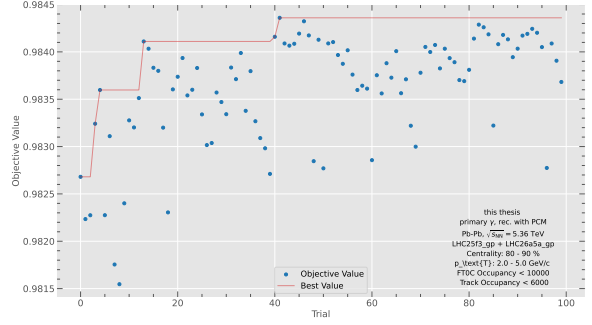
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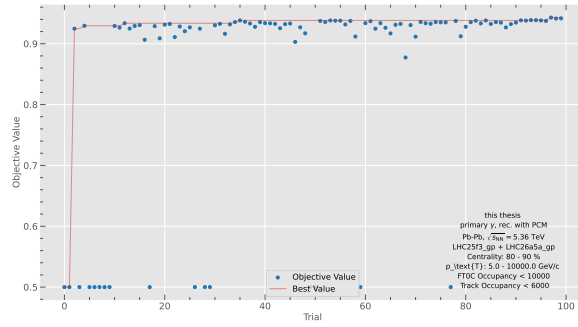
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

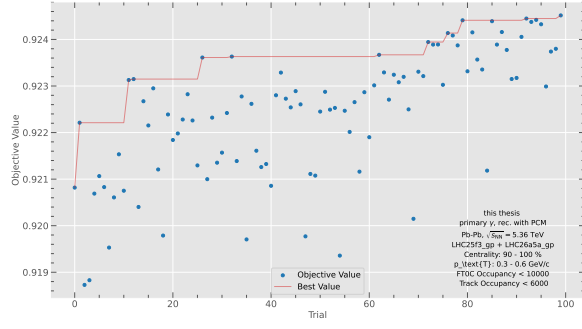


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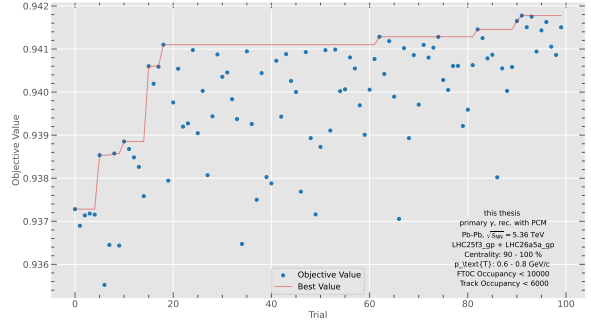


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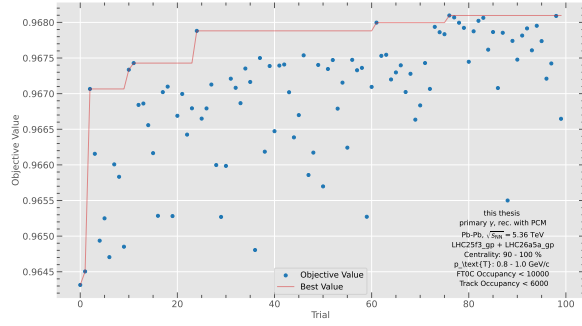
Figure 9: Hyperparameter optimisation history for all p_T bins in the centrality range 80-90 %



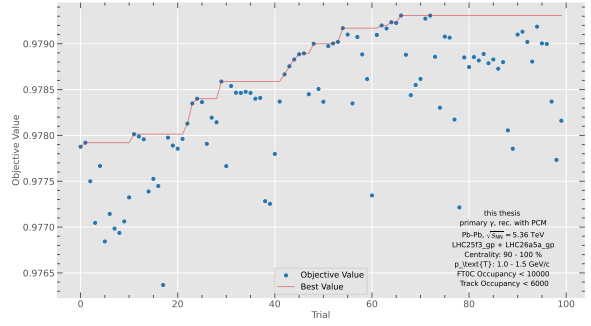
(a) 0.3-0.6 GeV/c



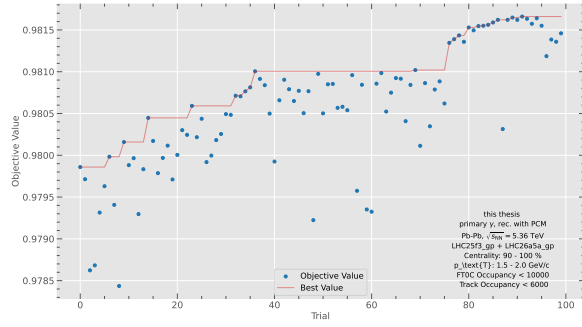
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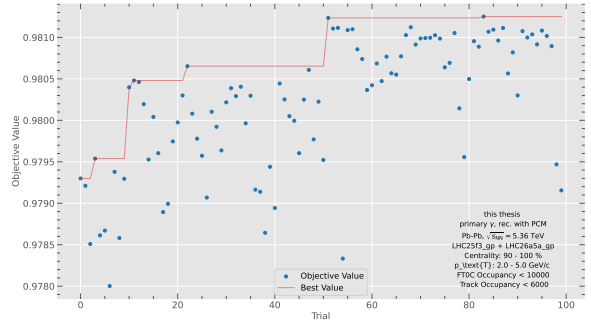
(c) 0.8-1.0 GeV/c



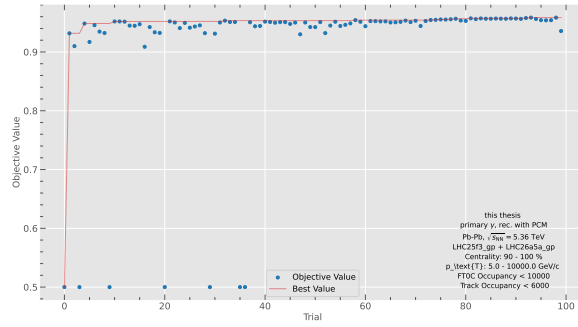
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c



(f) 2.0-5.0 GeV/c



(g) 5.0-10000.0 GeV/c

Figure 10: Hyperparameter optimisation history for all p_T bins in the centrality range 90-100 %

2 Hyperparameter evolution throughout the tuning process

2.1 Gamma, learning rate, maximum depth, minimum child weight, and number of estimators

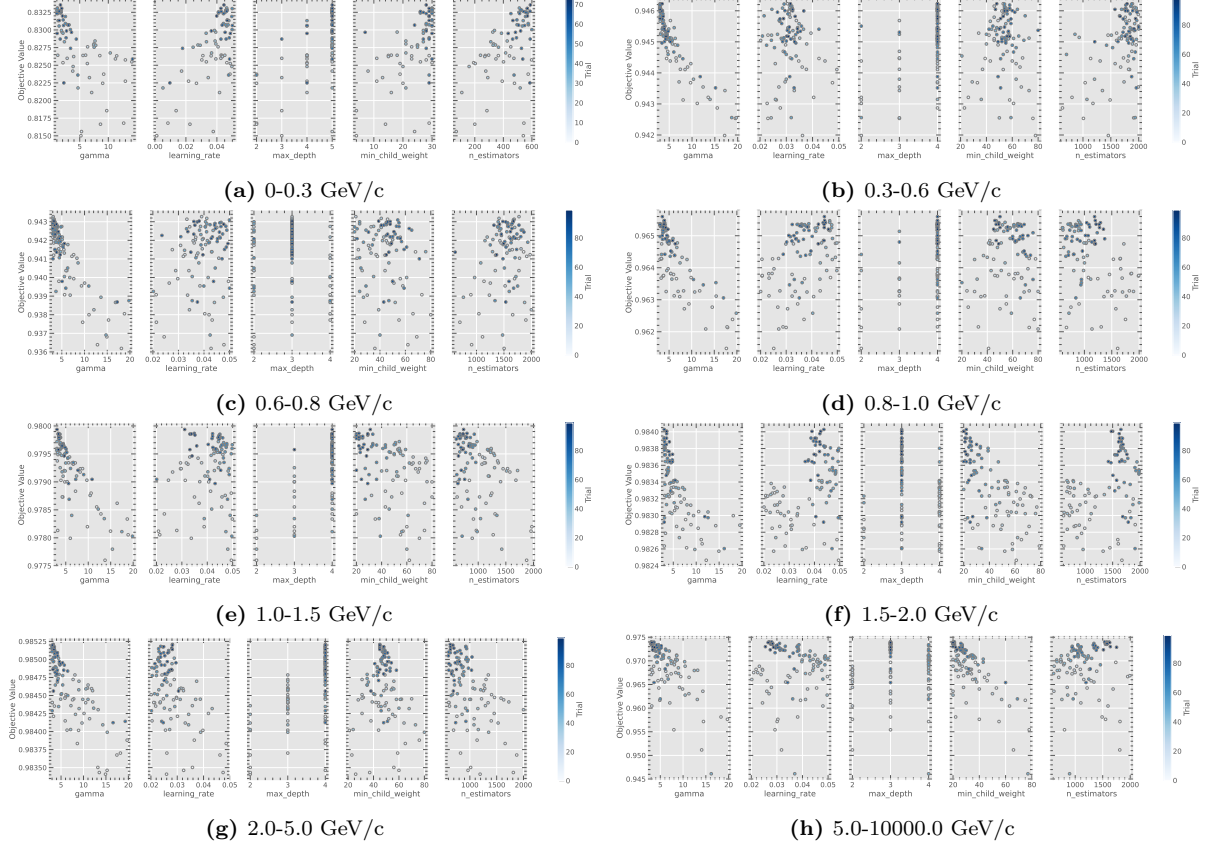


Figure 11: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 0-10 %

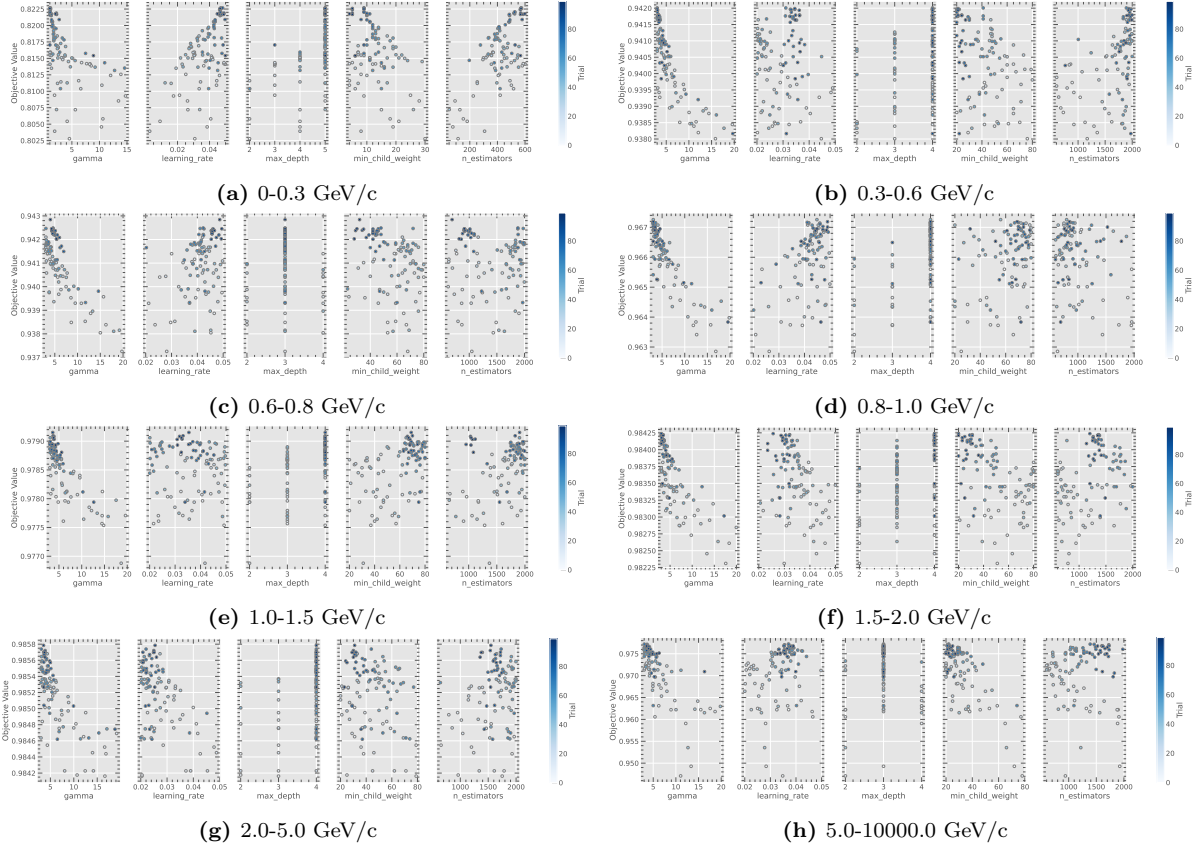


Figure 12: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 10-20 %

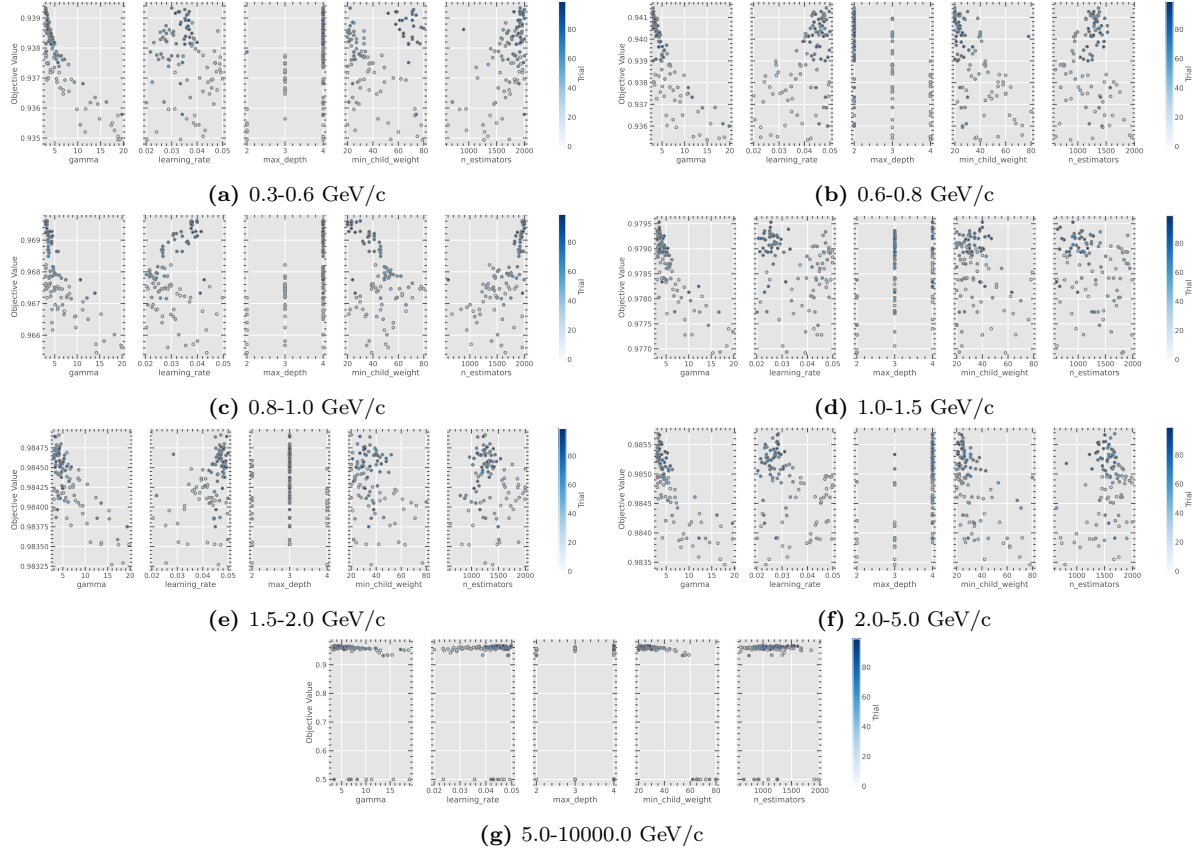


Figure 13: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 20-30 %

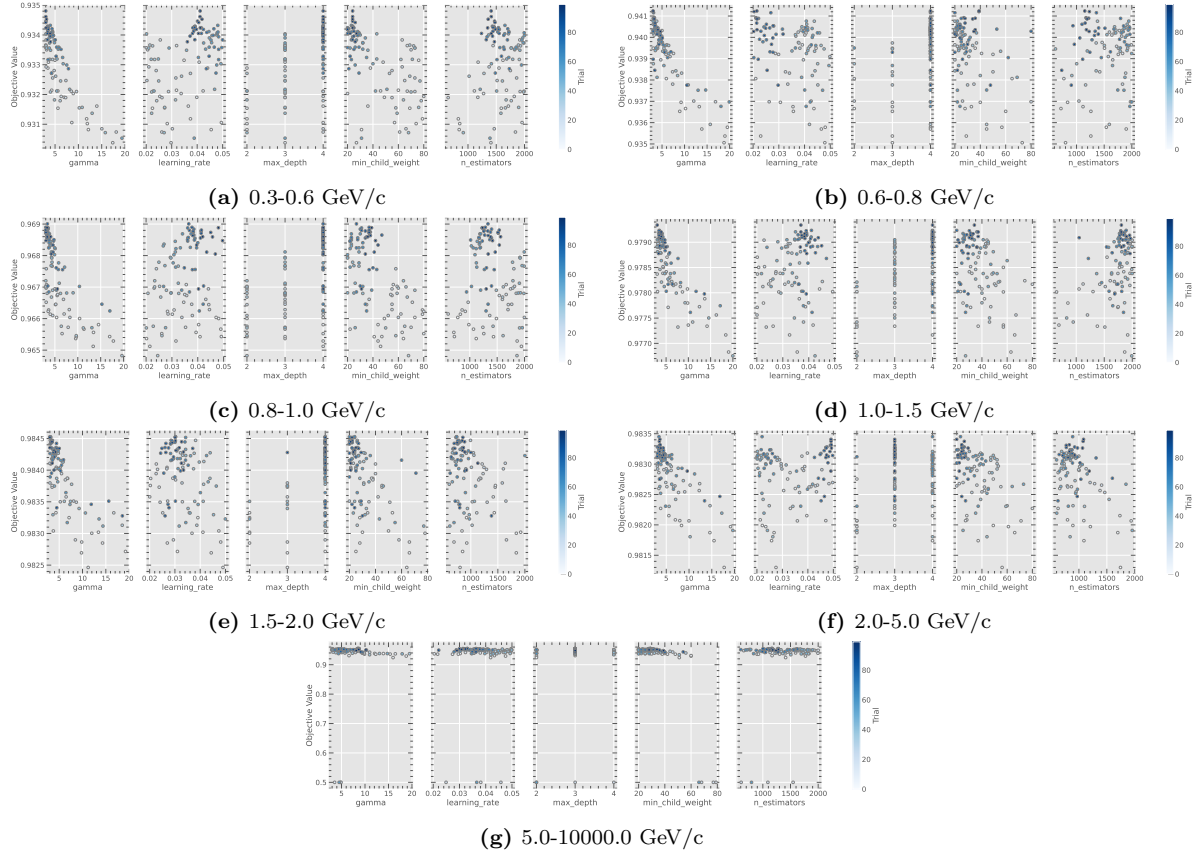


Figure 14: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 30-40 %

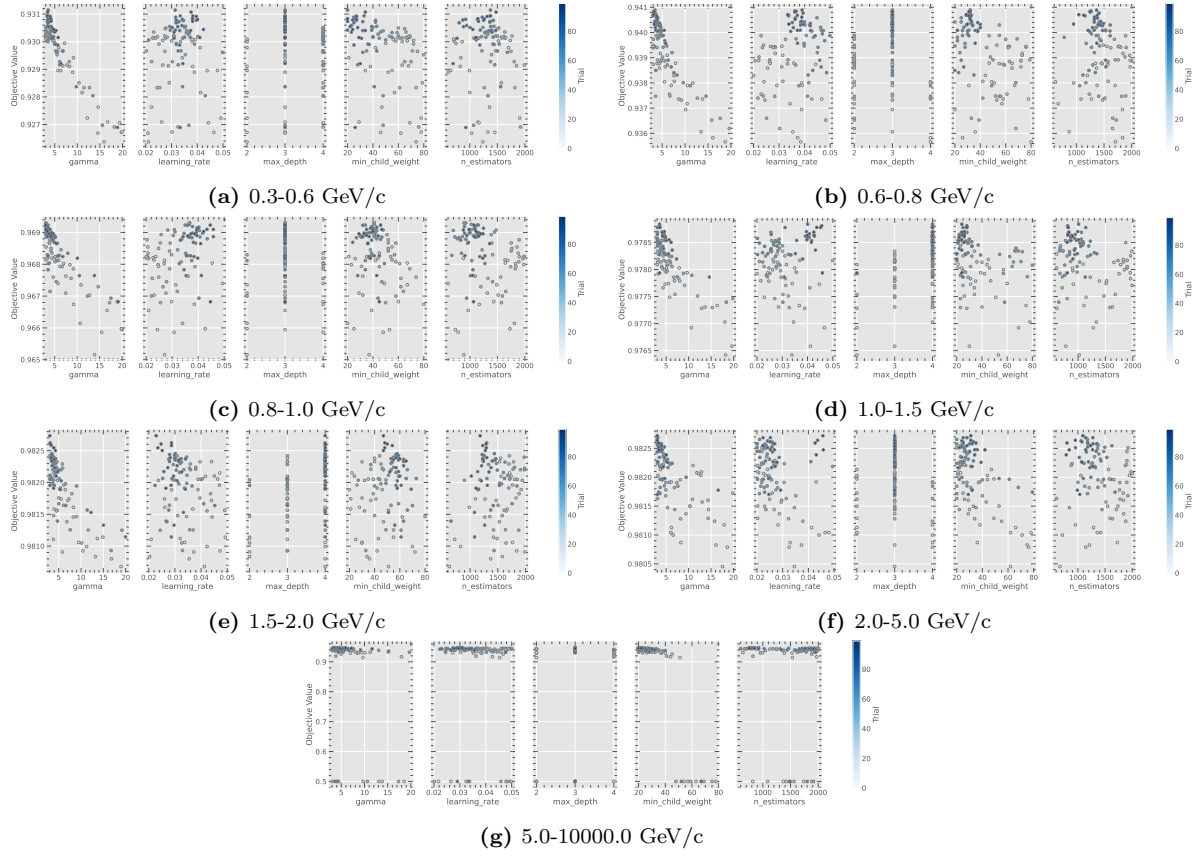


Figure 15: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 40-50 %

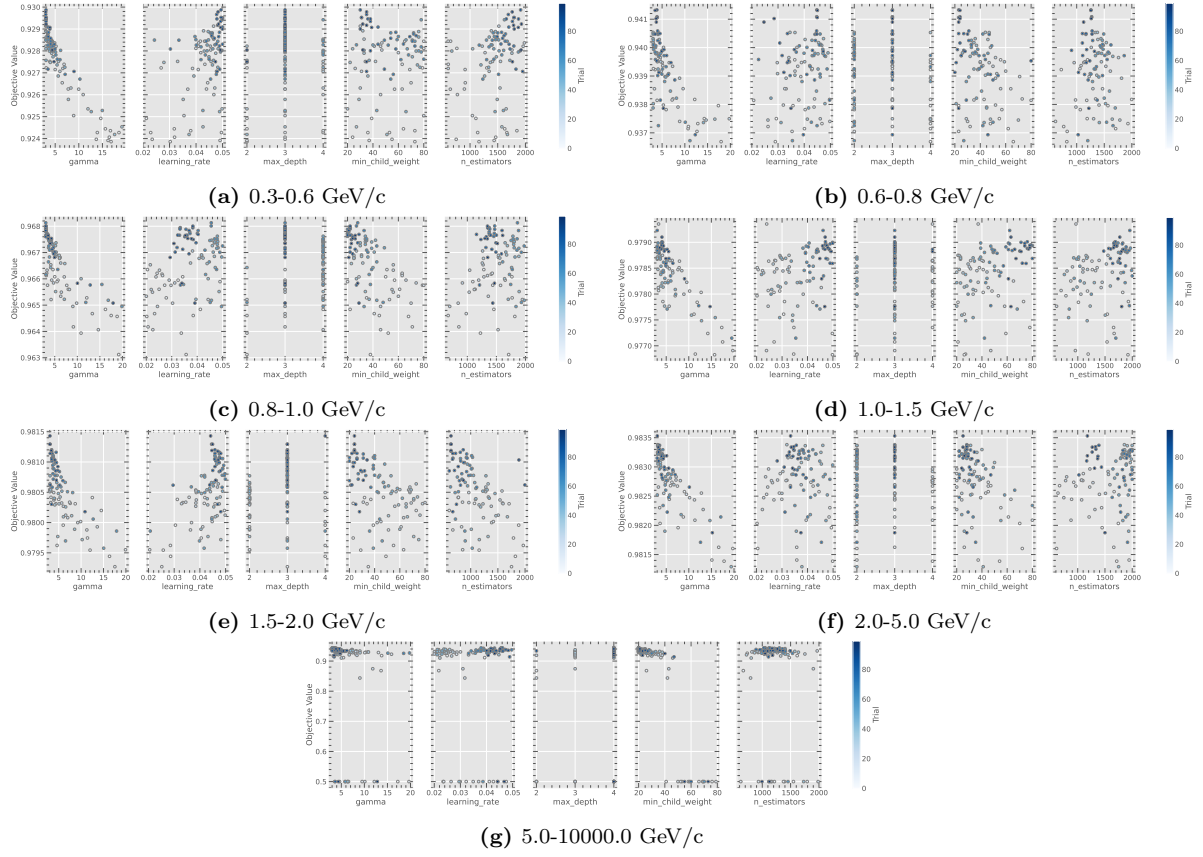


Figure 16: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 50-60 %

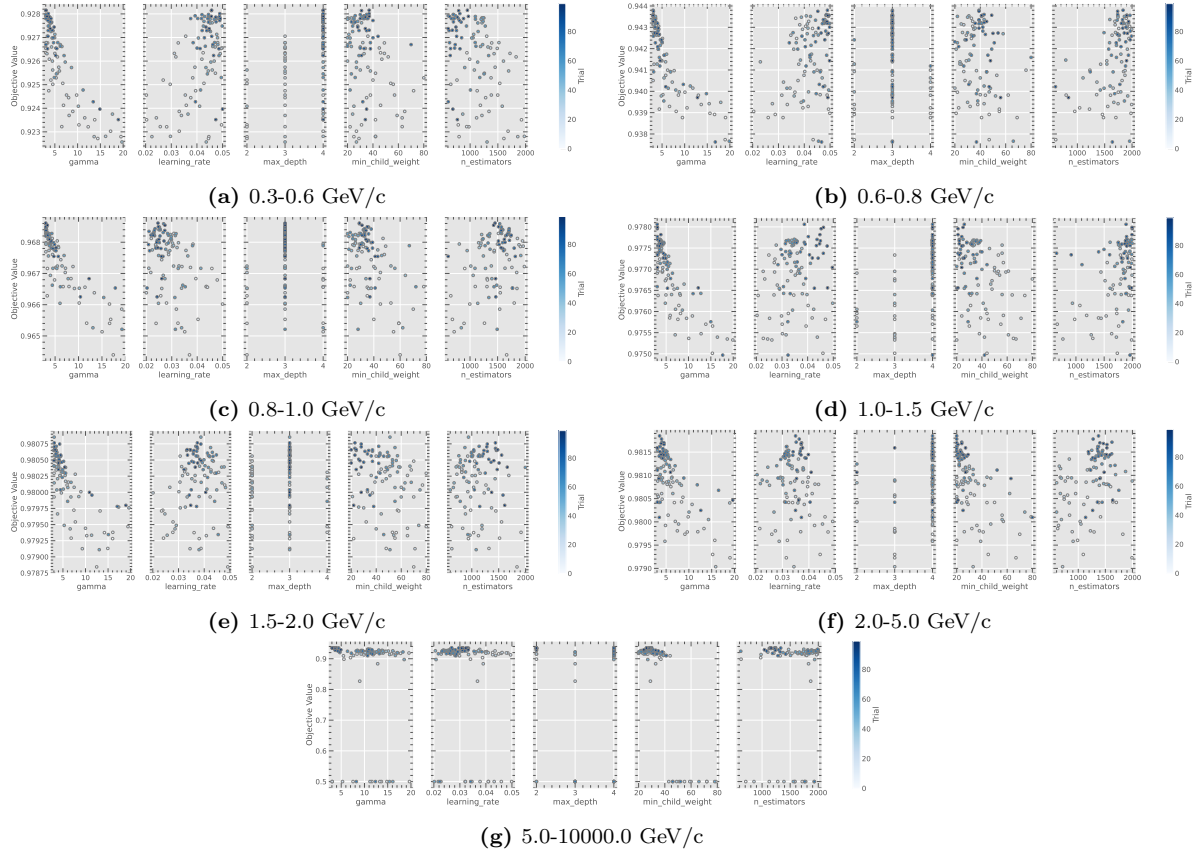


Figure 17: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 60-70 %

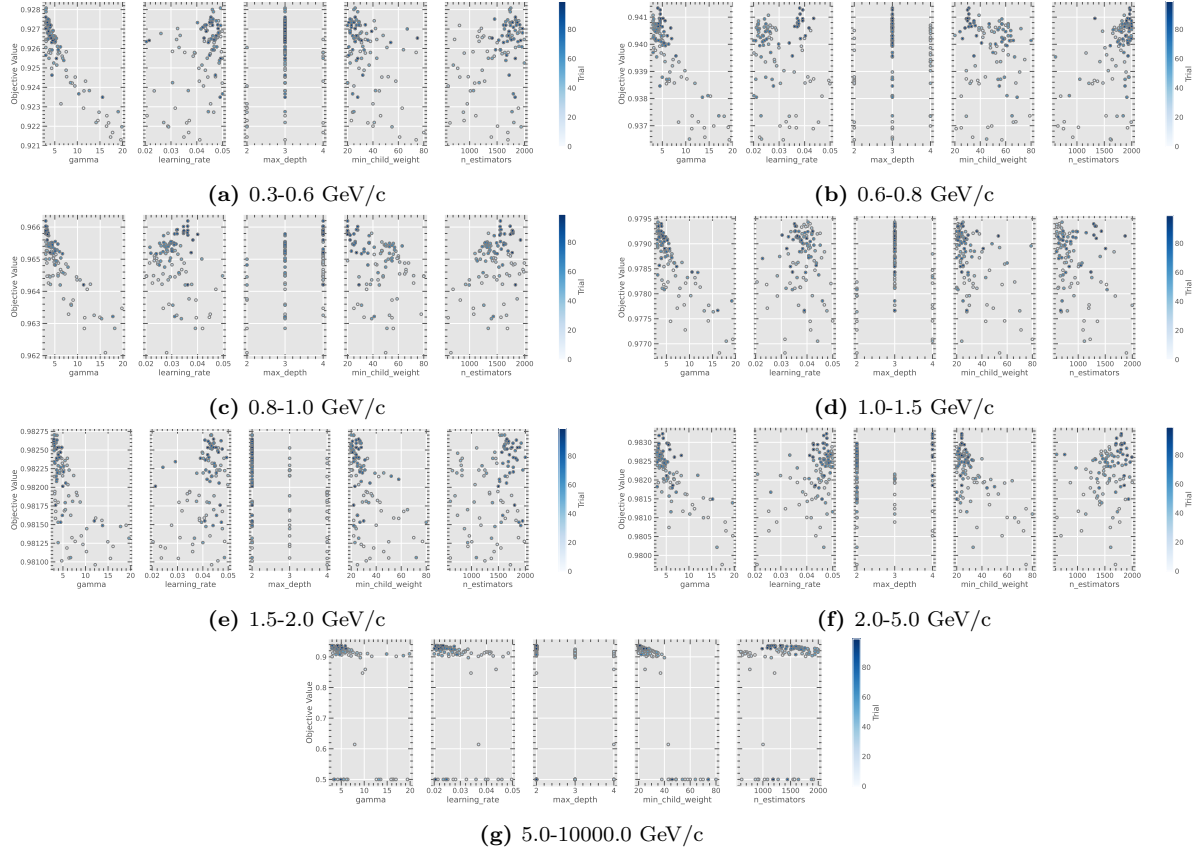


Figure 18: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 70-80 %

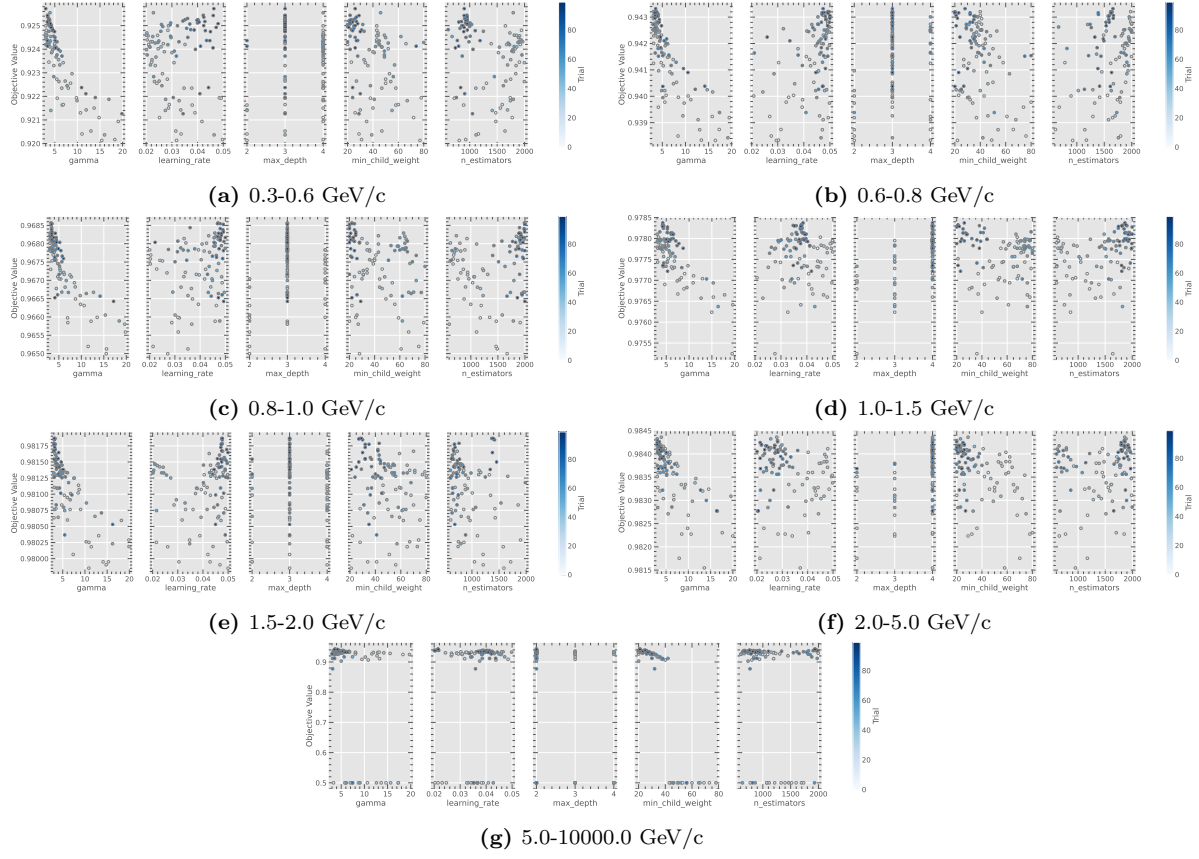


Figure 19: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 80-90 %

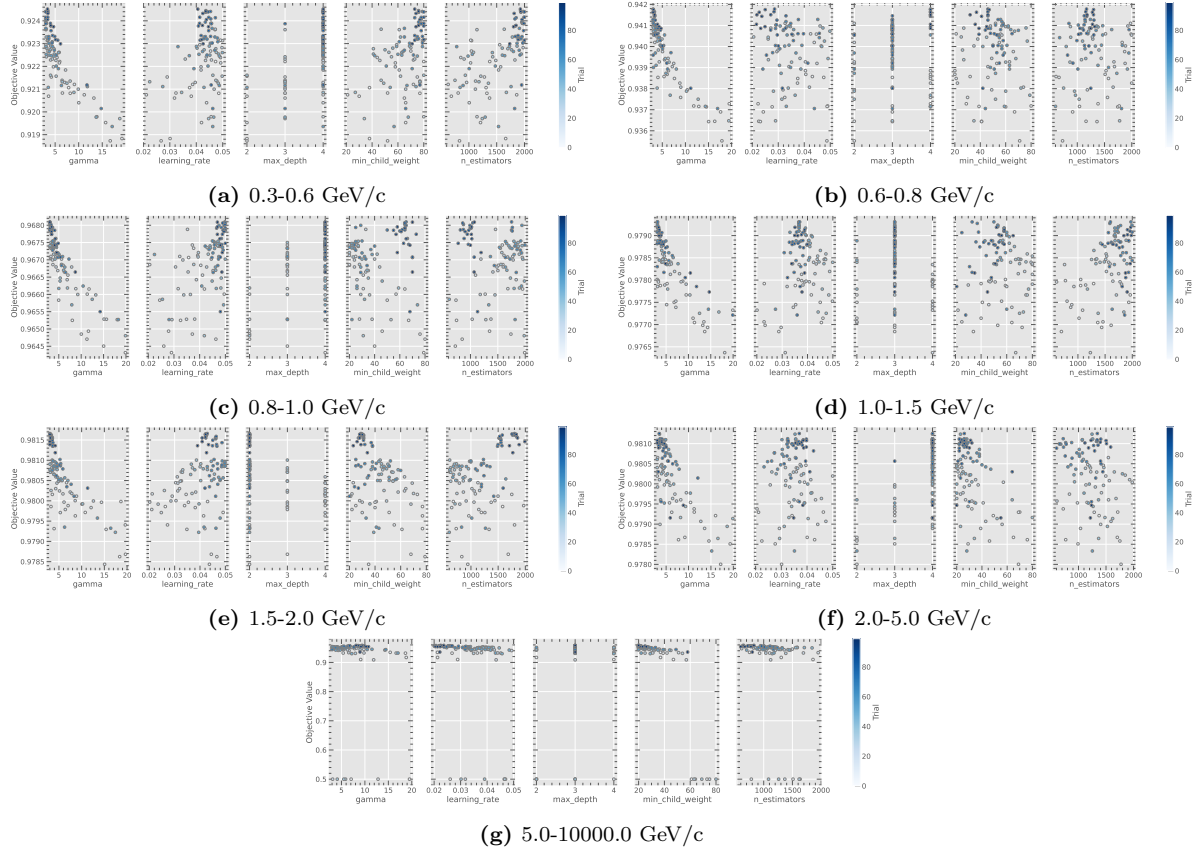


Figure 20: Hyperparameter evolution of `gamma`, `learning_rate`, `max_depth`, `min_child_weight`, and `n_estimators` throughout the tuning process for all p_T bins in the centrality range 90-100 %

2.2 Colsample by tree, Lasso and Ridge regularisation, class imbalance, and subsample

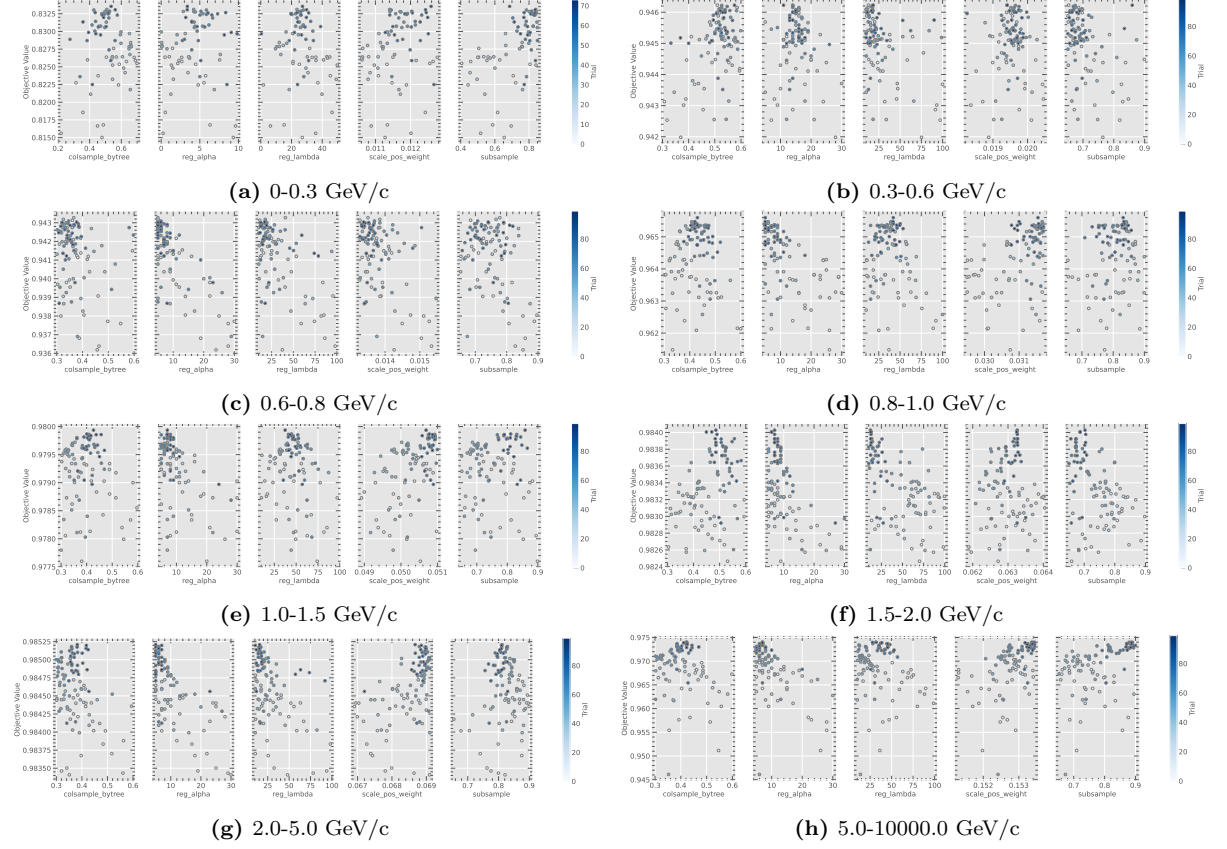


Figure 21: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 0-10 %

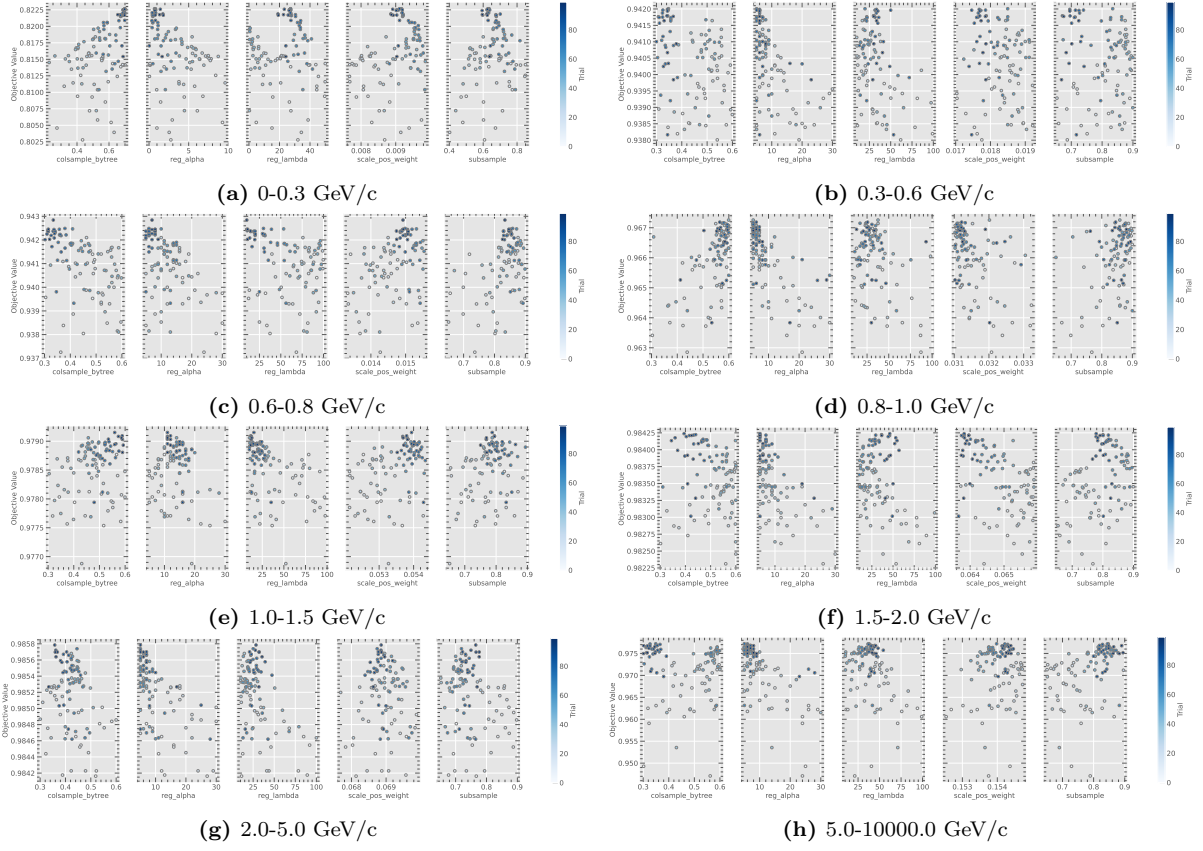


Figure 22: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 10-20 %

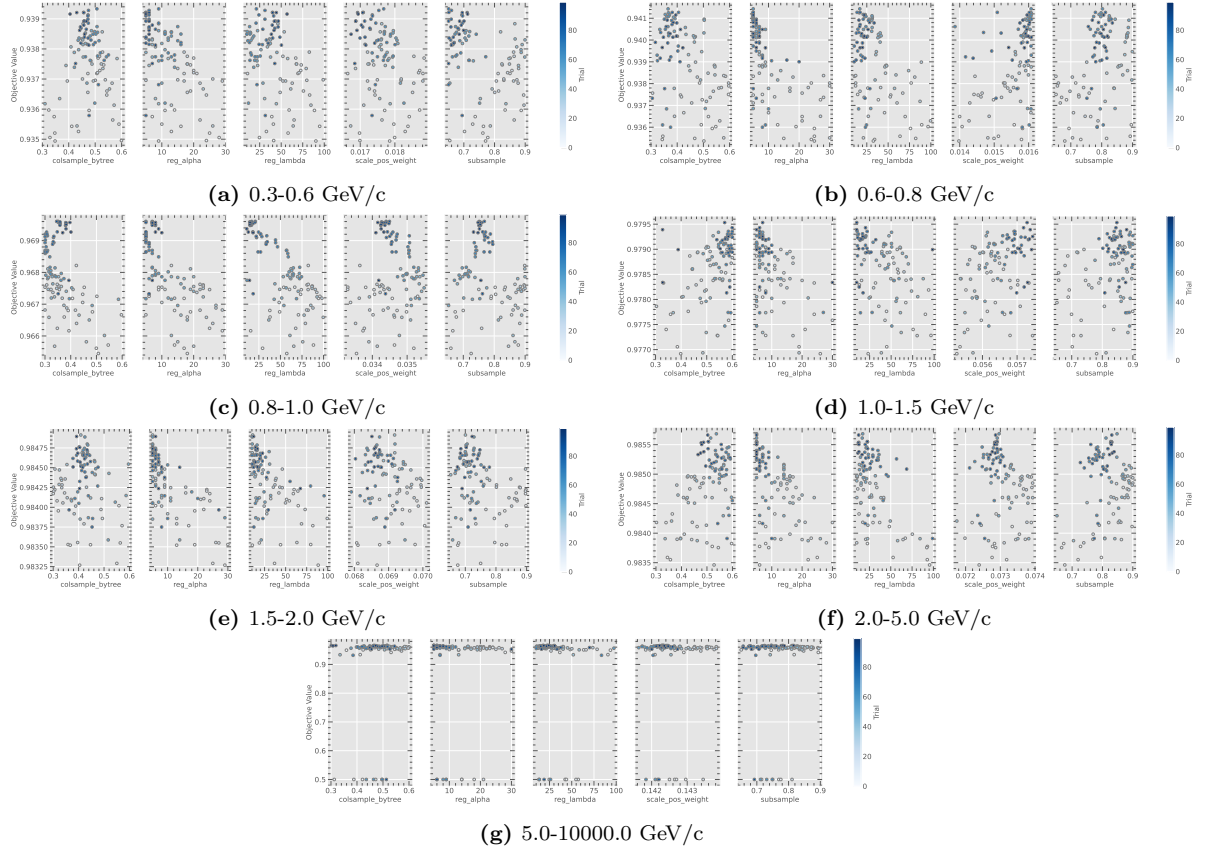


Figure 23: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 20-30 %

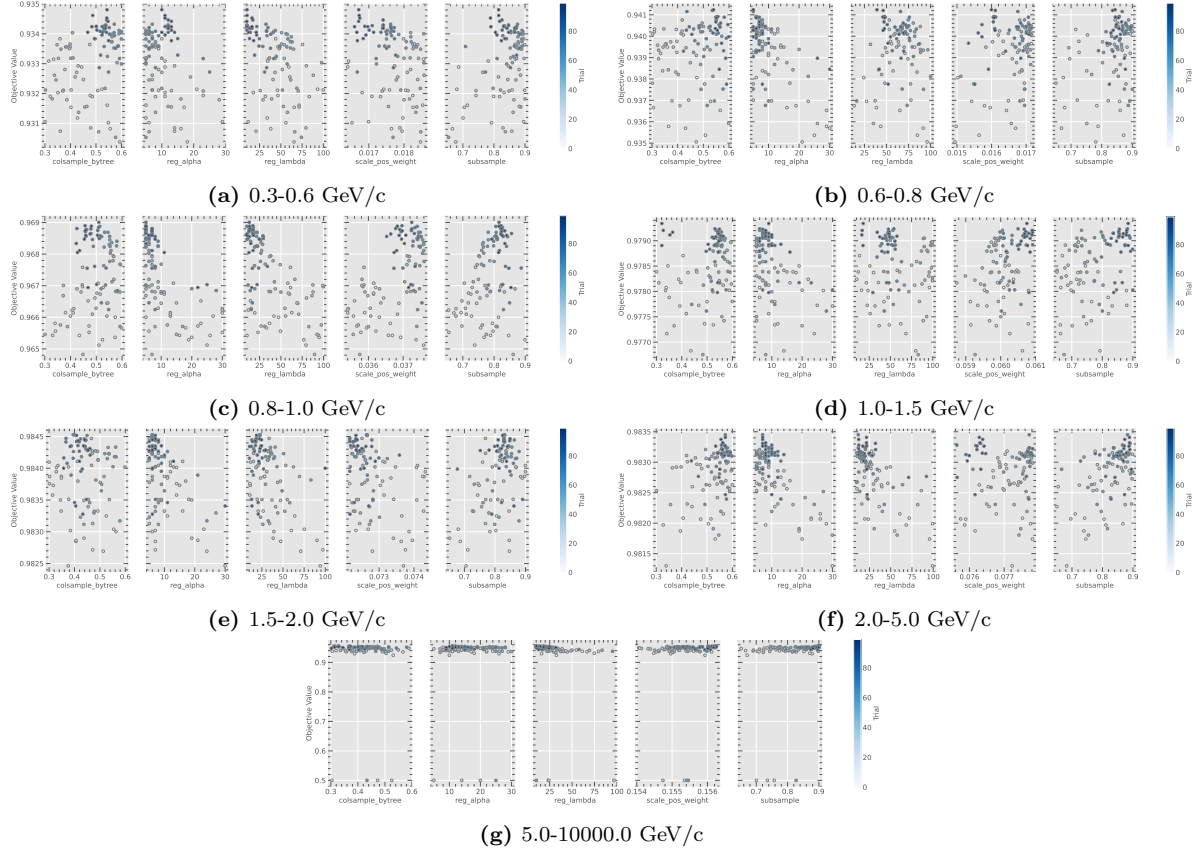


Figure 24: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 30-40 %

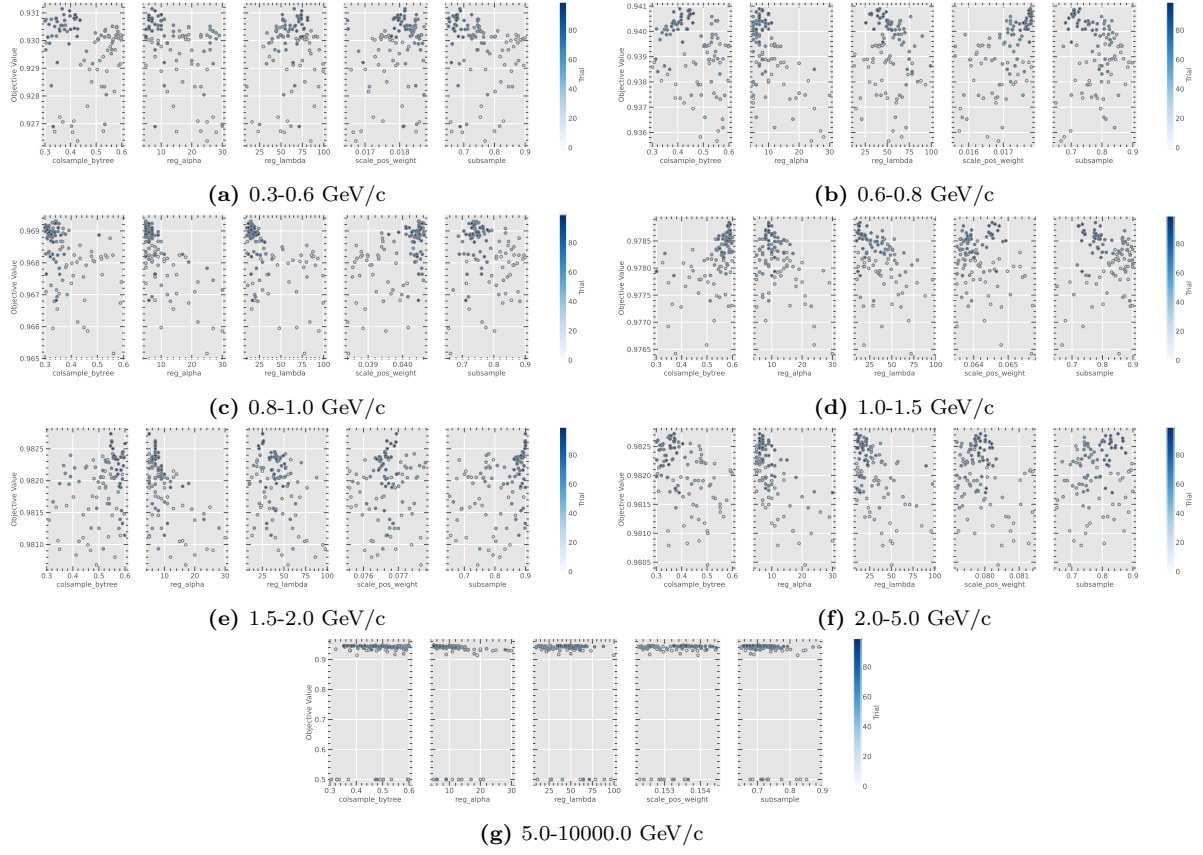


Figure 25: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 40-50 %

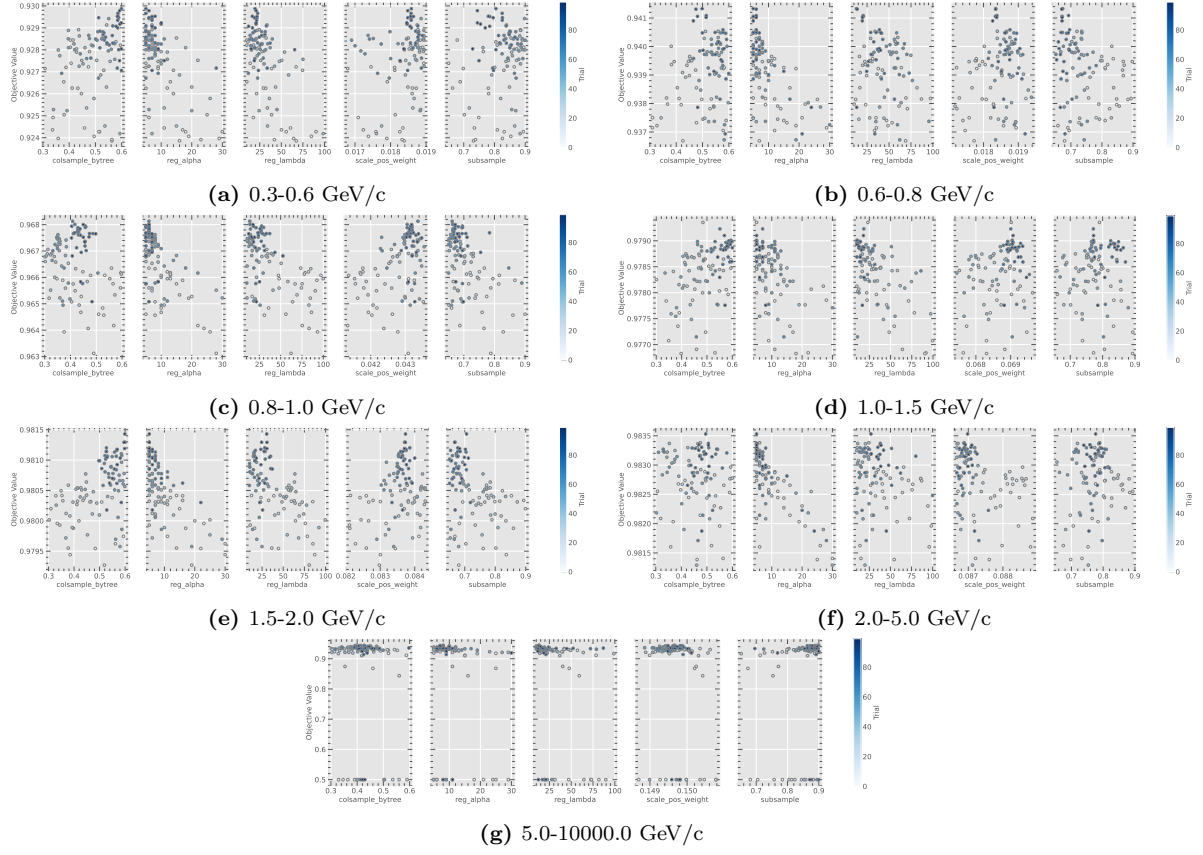


Figure 26: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 50-60 %

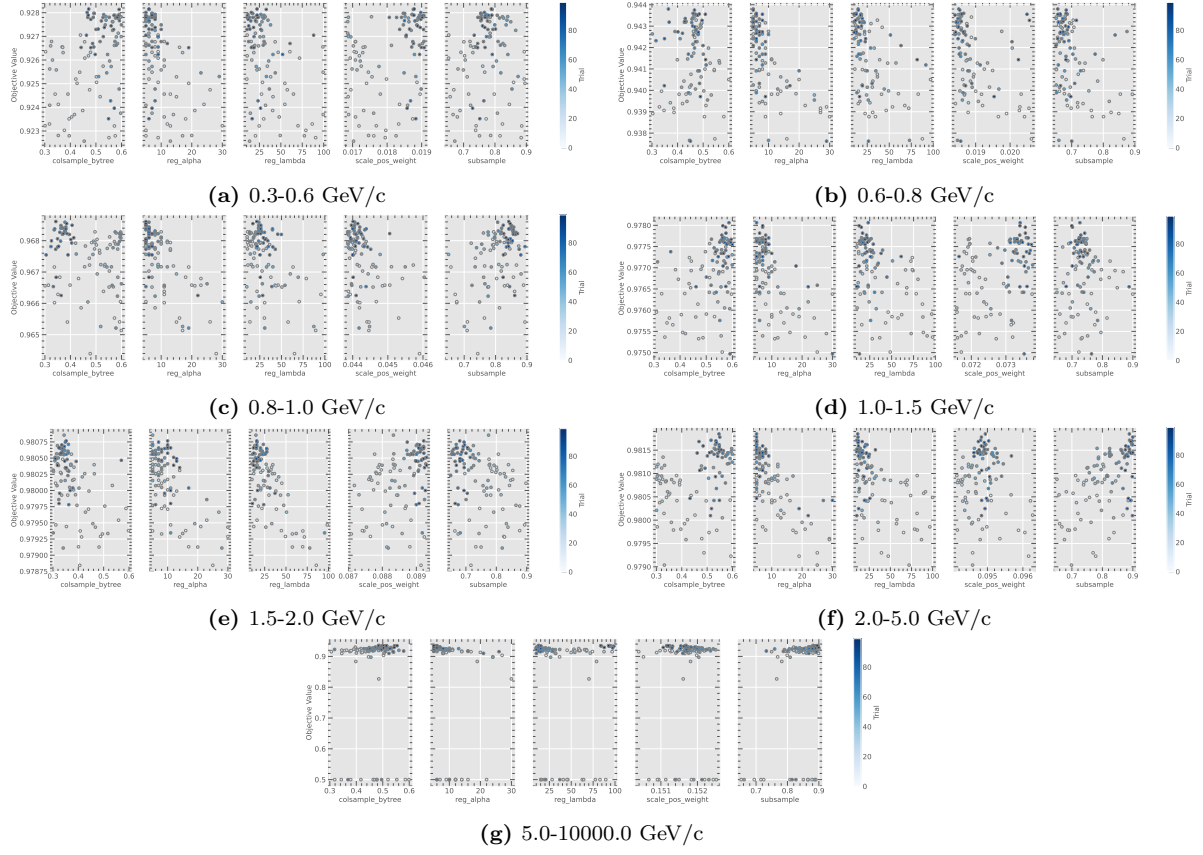


Figure 27: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 60-70 %

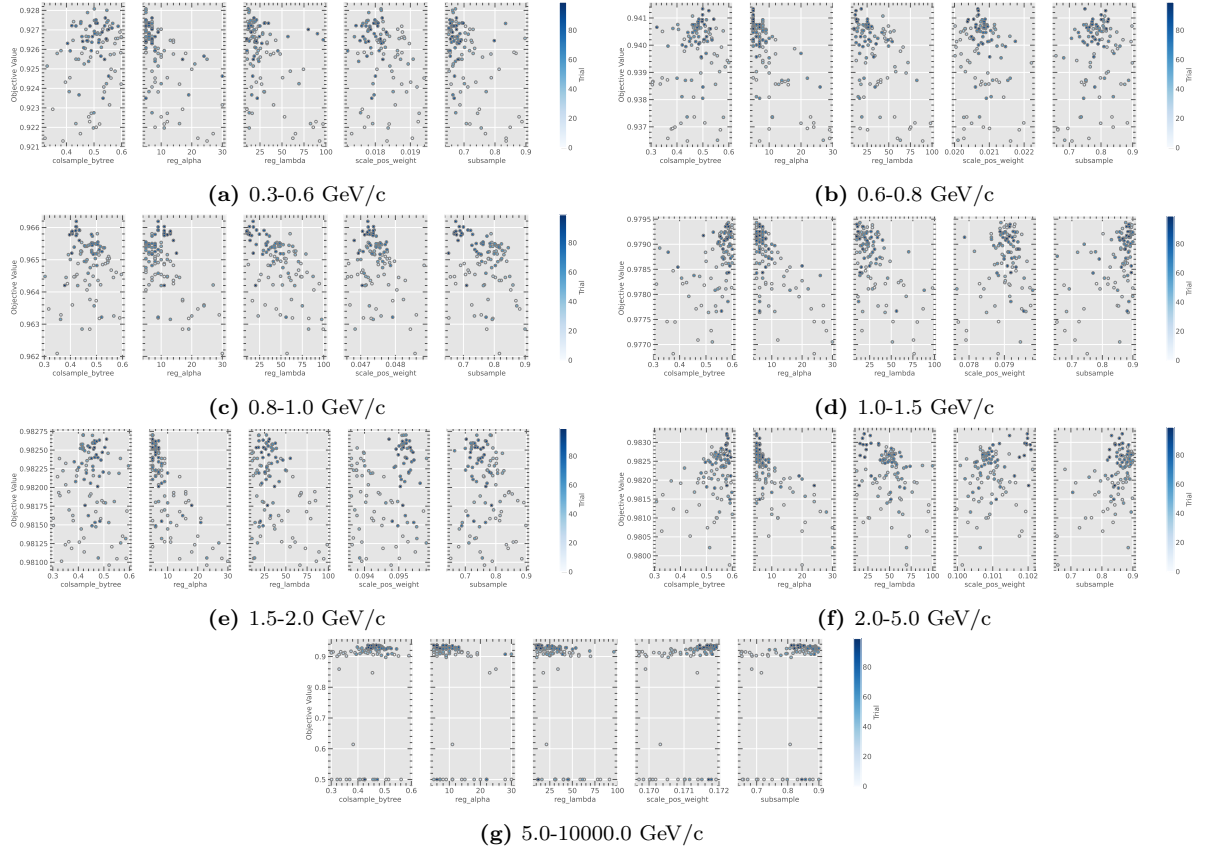


Figure 28: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 70-80 %

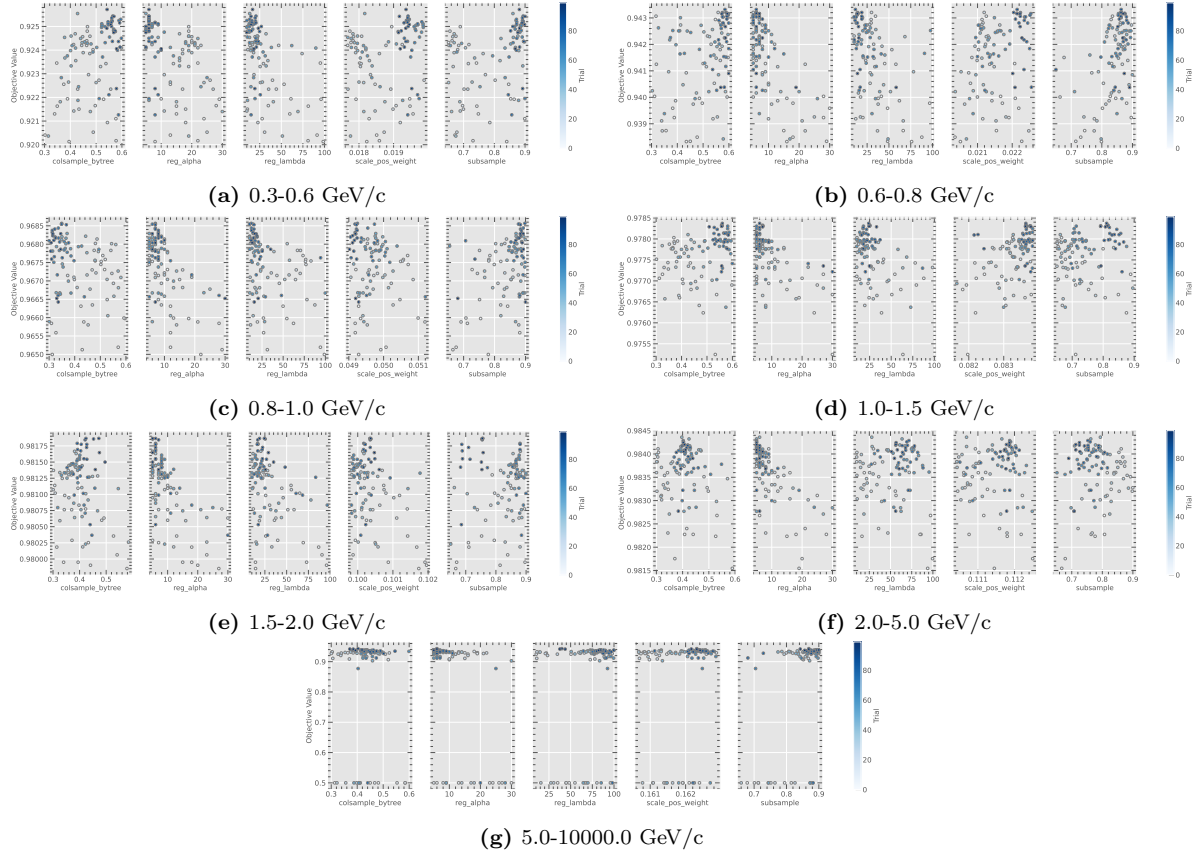


Figure 29: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 80-90 %

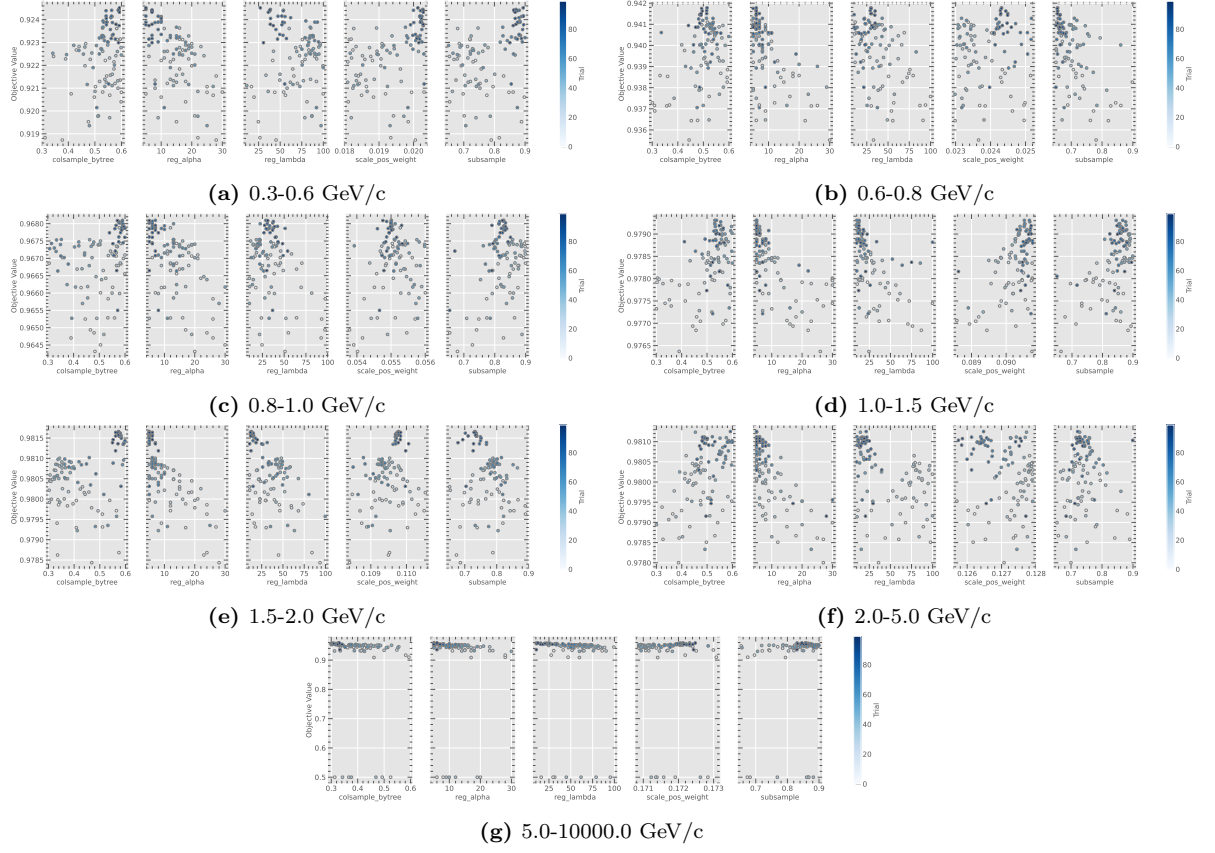


Figure 30: Hyperparameter evolution of `colsample_bytree`, `reg_alpha`, `reg_lambda`, `scale_pos_weight`, and `subsample` throughout the tuning process for all p_T bins in the centrality range 90-100 %

3 Hyperparameter importance

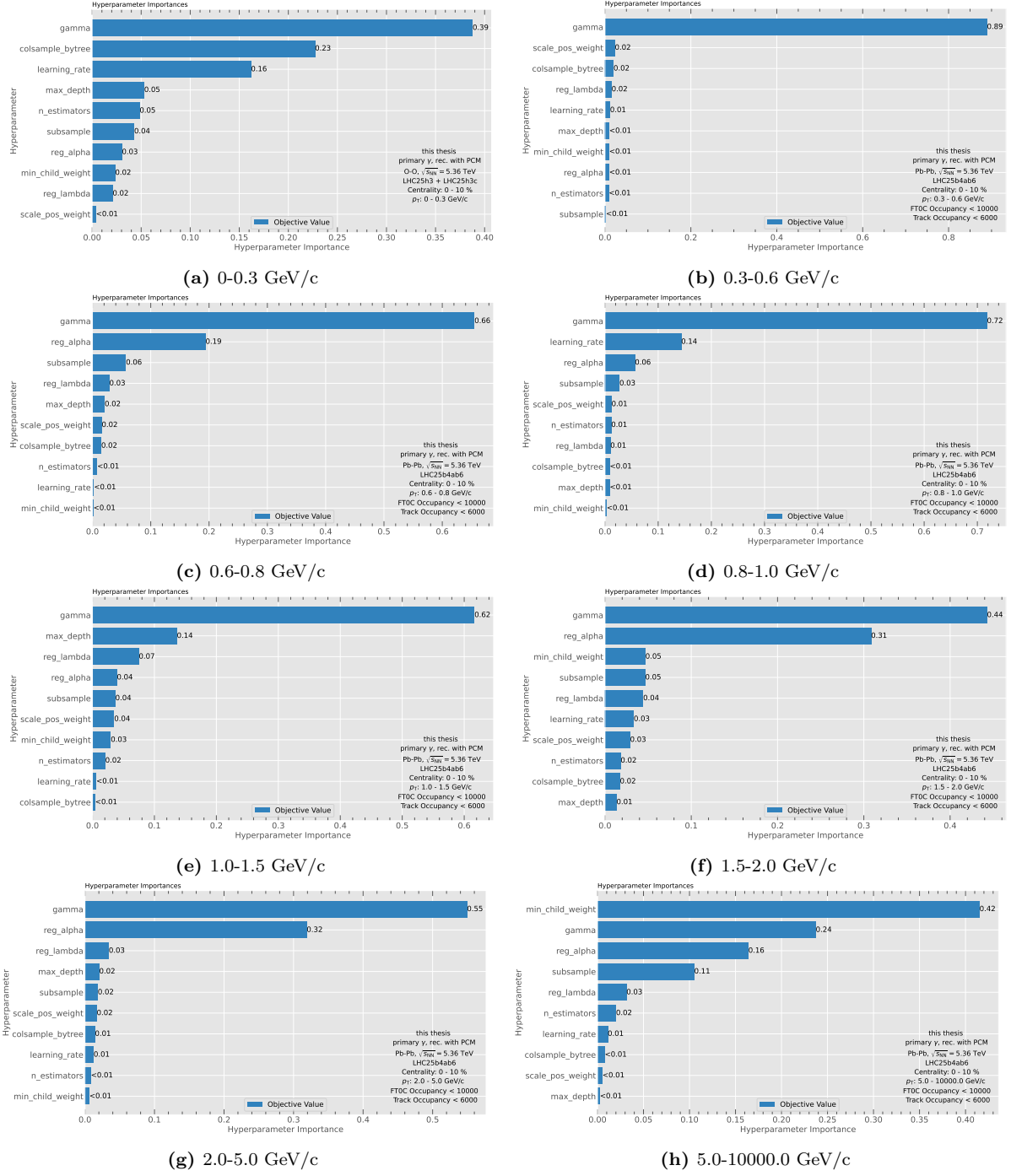
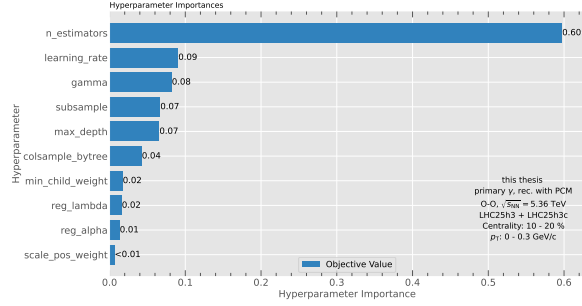
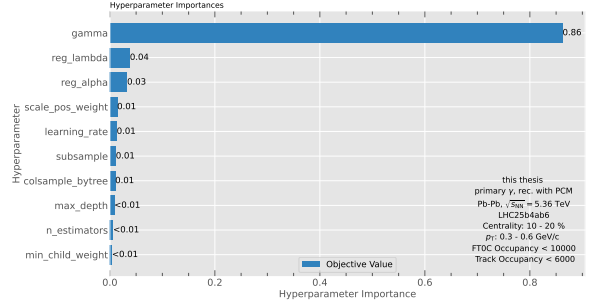


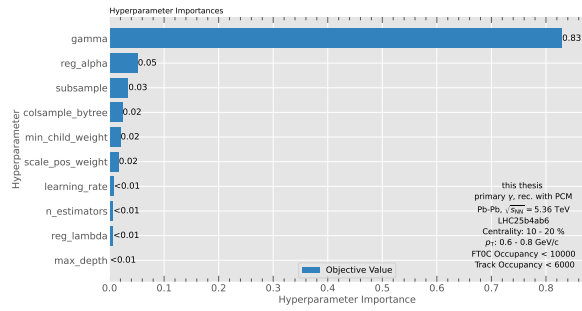
Figure 31: Hyperparameter importance for all p_T bins in the centrality range 0-10 %



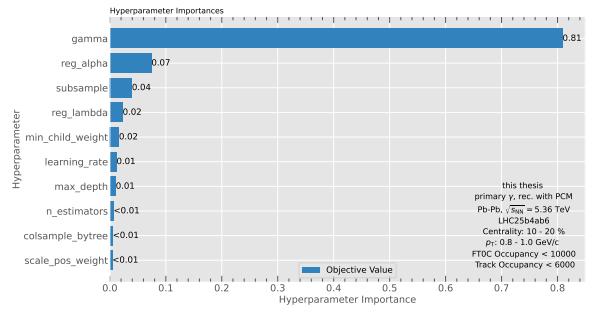
(a) 0-0.3 GeV/c



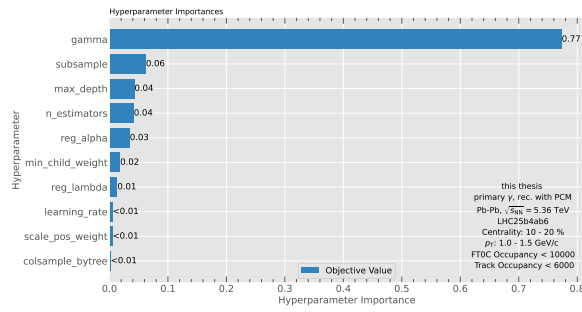
(b) 0.3-0.6 GeV/c



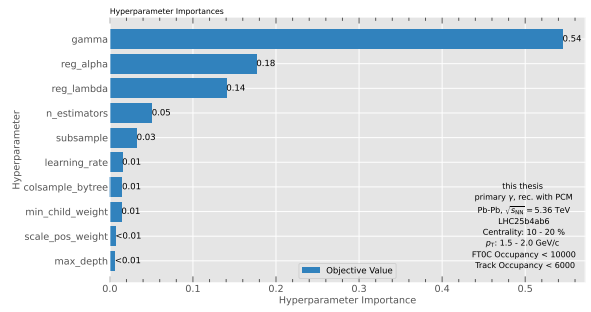
(c) 0.6-0.8 GeV/c



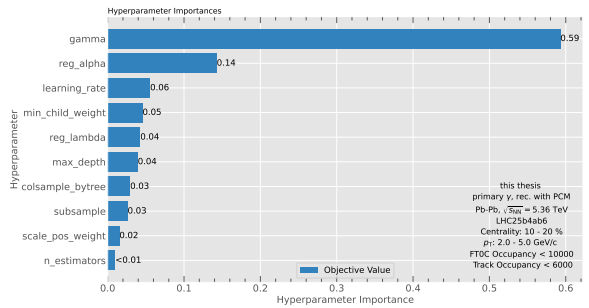
(d) 0.8-1.0 GeV/c



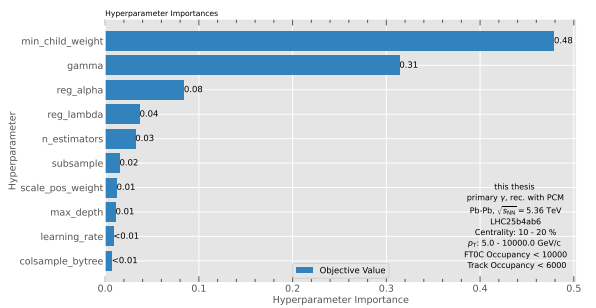
(e) 1.0-1.5 GeV/c



(f) 1.5-2.0 GeV/c

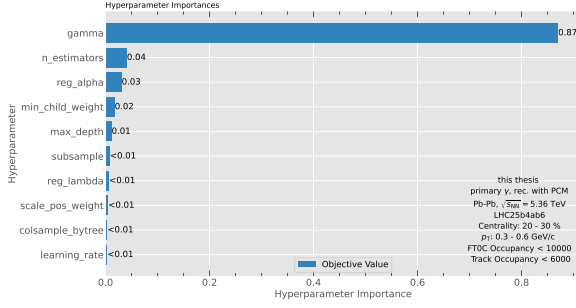


(g) 2.0-5.0 GeV/c

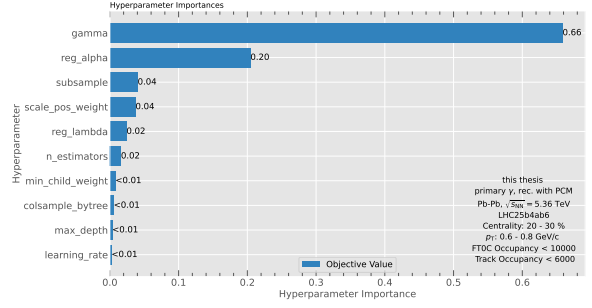


(h) 5.0-10000.0 GeV/c

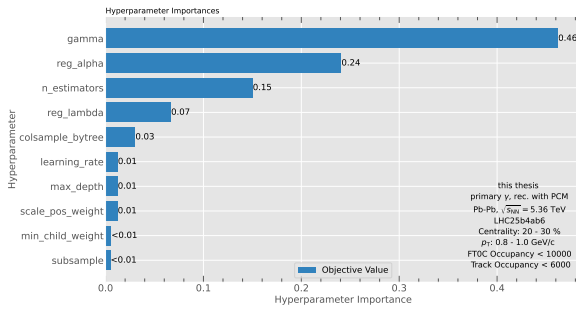
Figure 32: Hyperparameter importance for all p_T bins in the centrality range 10-20 %



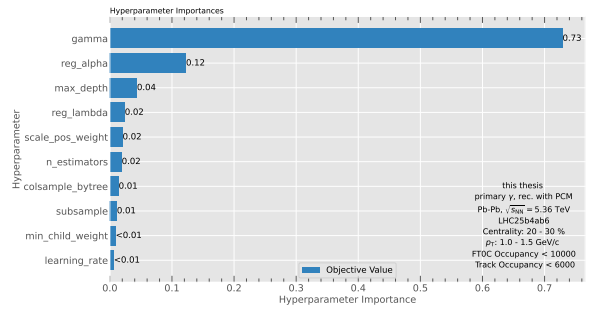
(a) 0.3-0.6 GeV/c



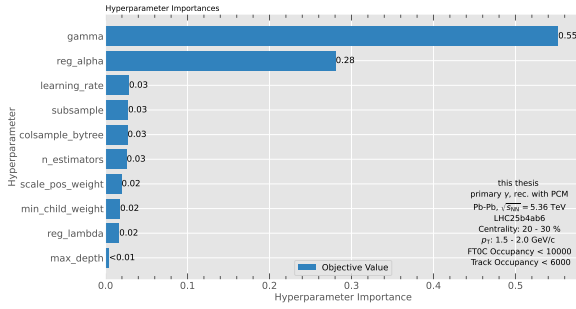
(b) 0.6-0.8 GeV/c



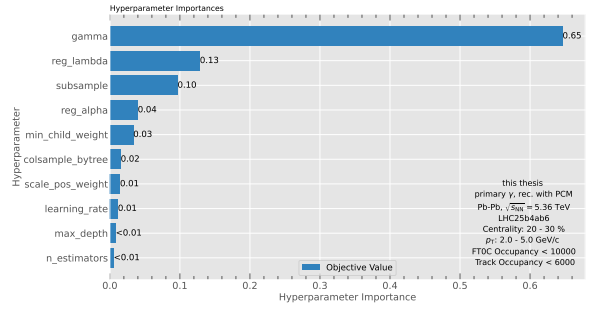
(c) 0.8-1.0 GeV/c



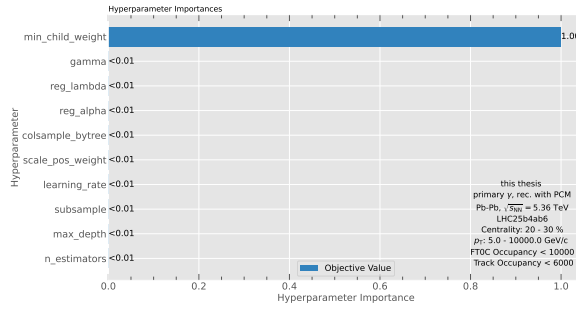
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

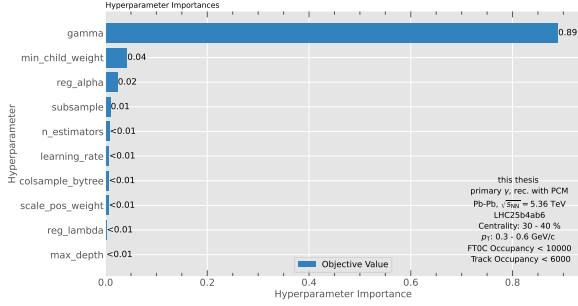


(f) 2.0-5.0 GeV/c

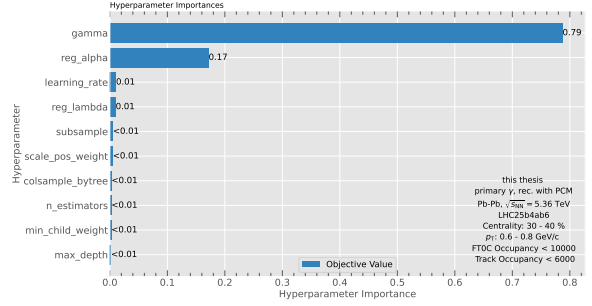


(g) 5.0-10000.0 GeV/c

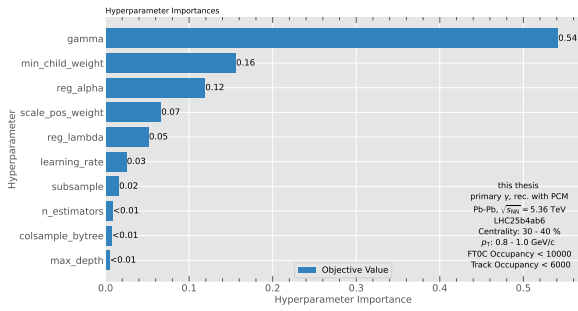
Figure 33: Hyperparameter importance for all p_T bins in the centrality range 20-30 %



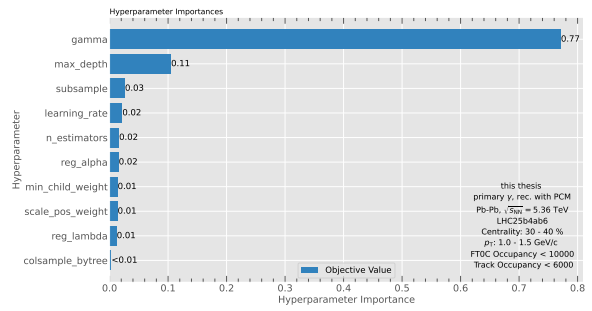
(a) 0.3-0.6 GeV/c



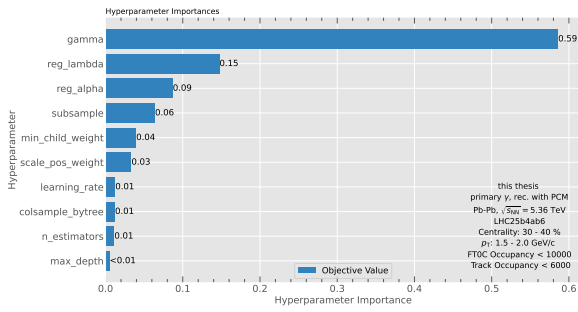
(b) 0.6-0.8 GeV/c



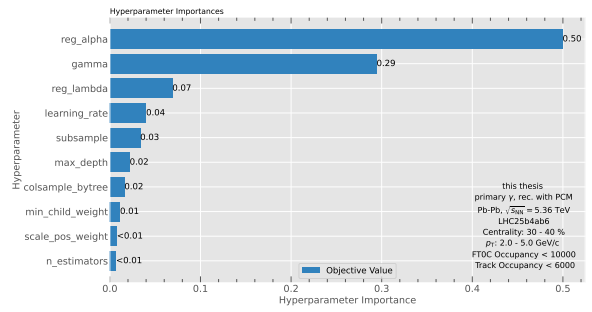
(c) 0.8-1.0 GeV/c



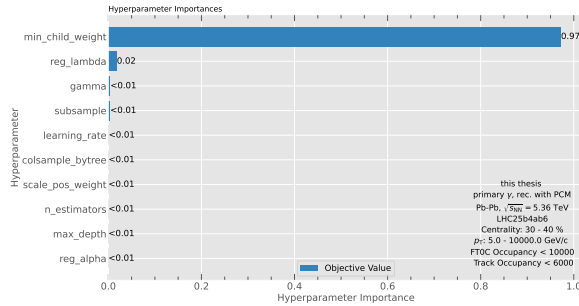
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

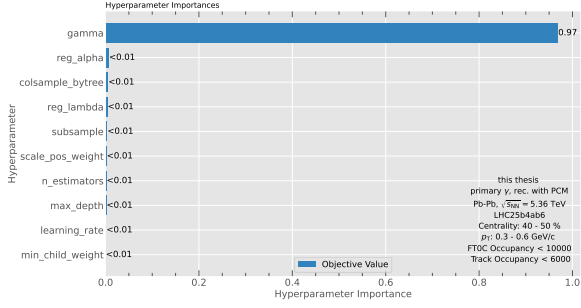


(f) 2.0-5.0 GeV/c

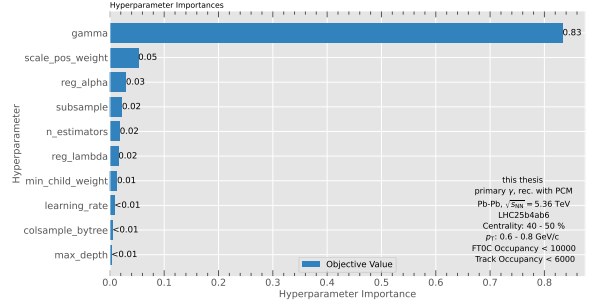


(g) 5.0-10000.0 GeV/c

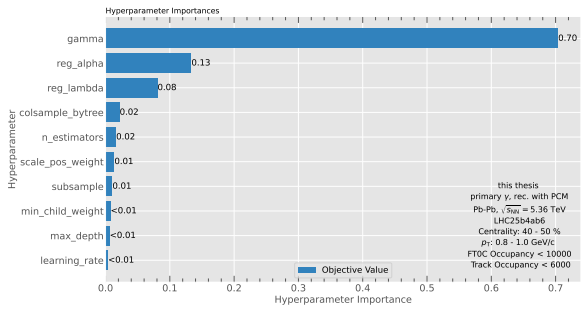
Figure 34: Hyperparameter importance for all p_T bins in the centrality range 30-40 %



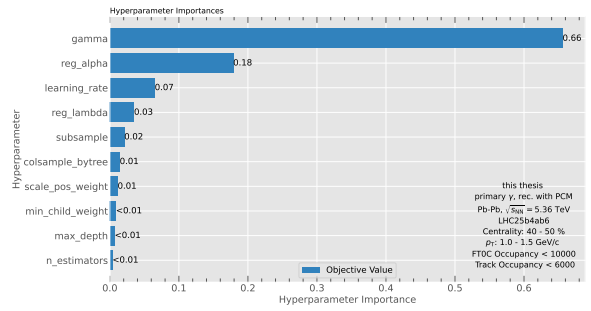
(a) 0.3-0.6 GeV/c



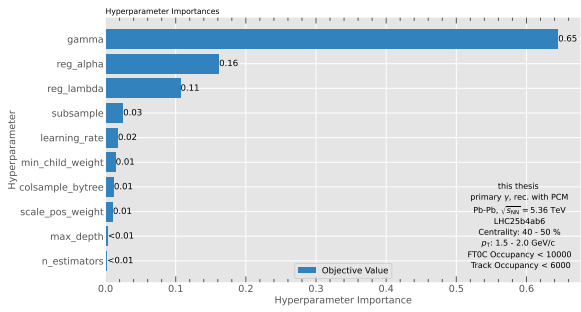
(b) 0.6-0.8 GeV/c



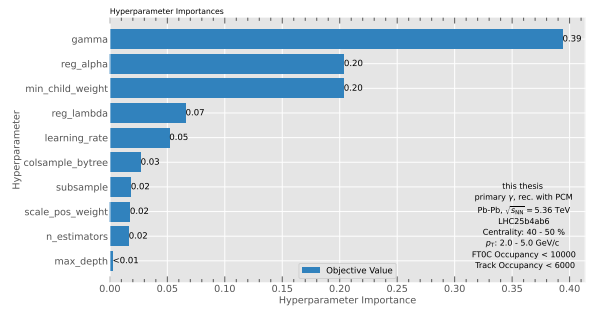
(c) 0.8-1.0 GeV/c



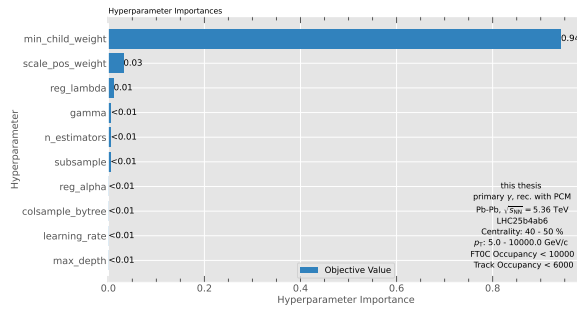
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

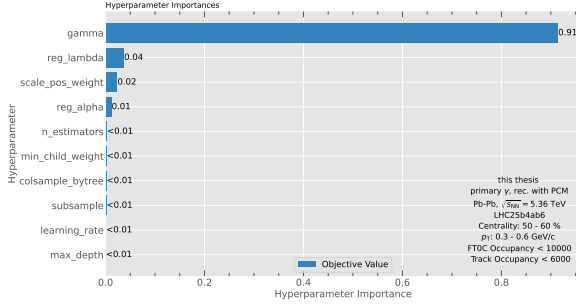


(f) 2.0-5.0 GeV/c

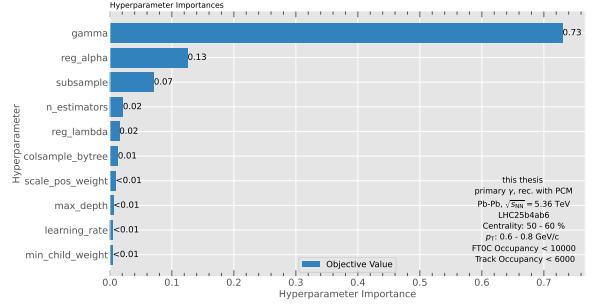


(g) 5.0-10000.0 GeV/c

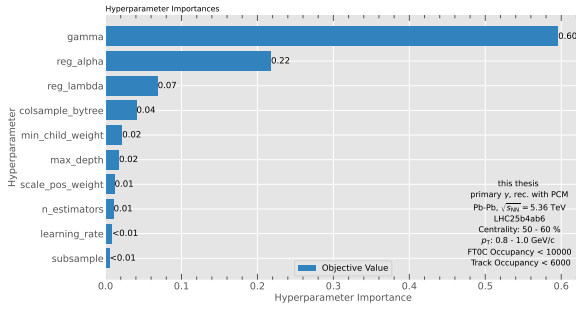
Figure 35: Hyperparameter importance for all p_T bins in the centrality range 40-50 %



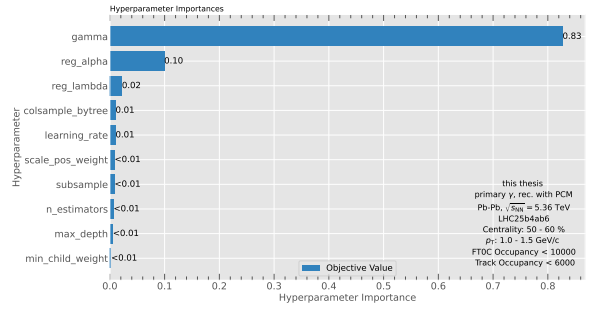
(a) 0.3-0.6 GeV/c



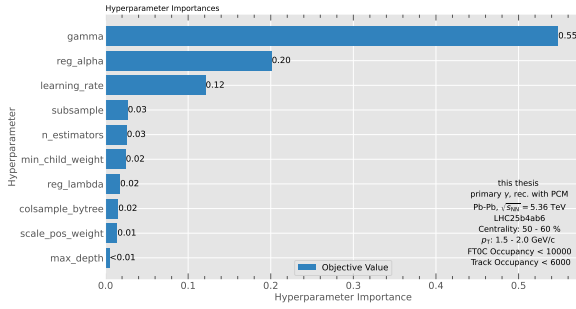
(b) 0.6-0.8 GeV/c



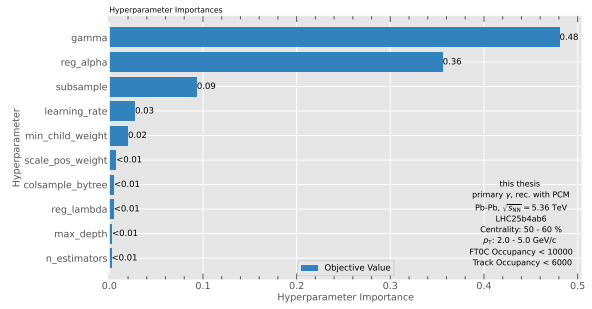
(c) 0.8-1.0 GeV/c



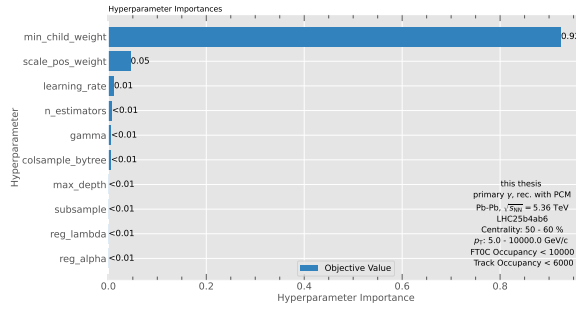
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

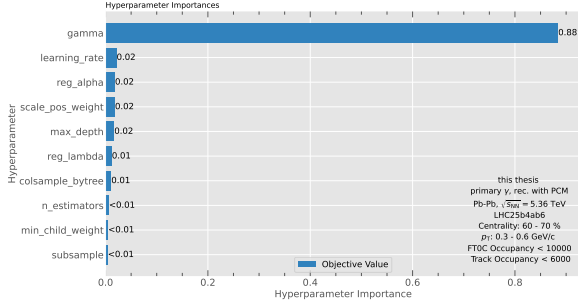


(f) 2.0-5.0 GeV/c

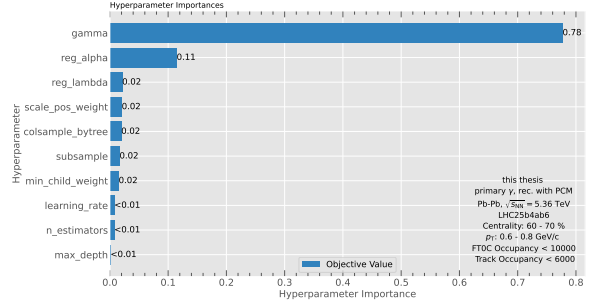


(g) 5.0-10000.0 GeV/c

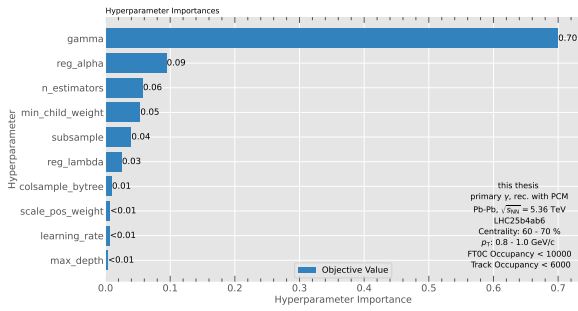
Figure 36: Hyperparameter importance for all p_T bins in the centrality range 50-60 %



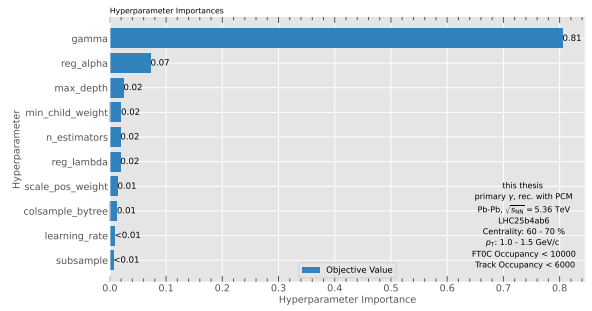
(a) 0.3-0.6 GeV/c



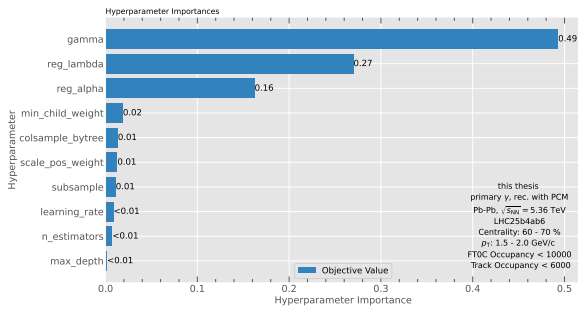
(b) 0.6-0.8 GeV/c



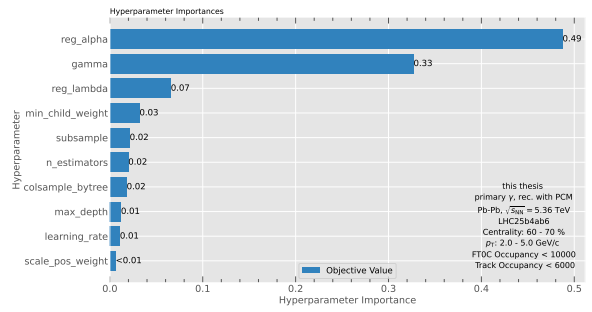
(c) 0.8-1.0 GeV/c



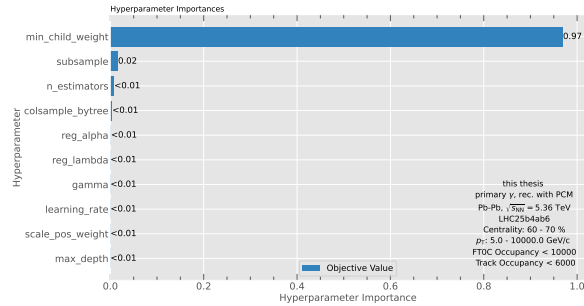
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

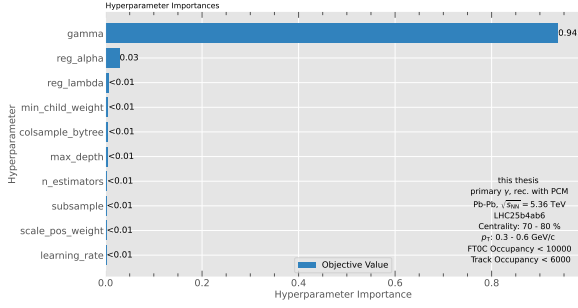


(f) 2.0-5.0 GeV/c

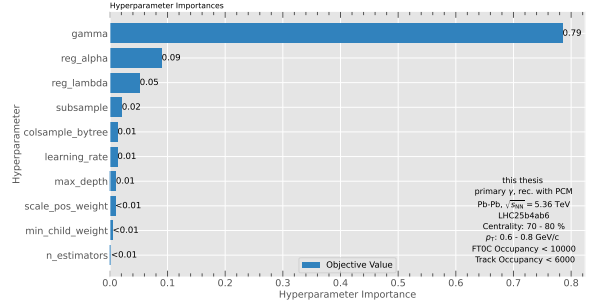


(g) 5.0-10000.0 GeV/c

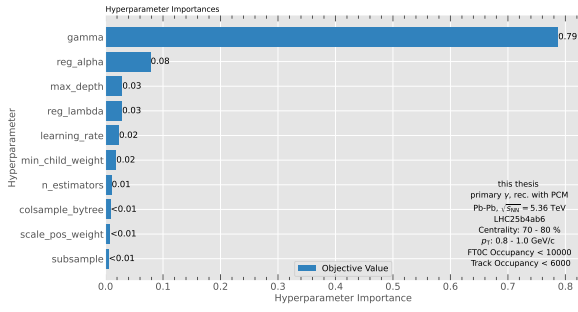
Figure 37: Hyperparameter importance for all p_T bins in the centrality range 60-70 %



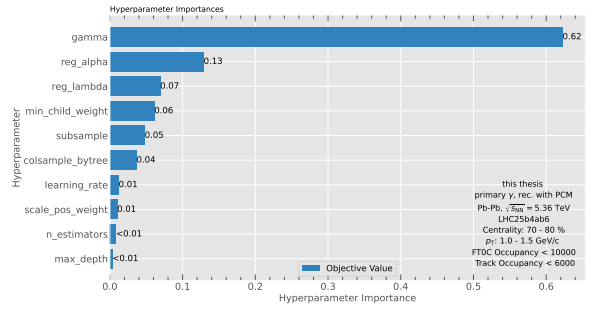
(a) 0.3-0.6 GeV/c



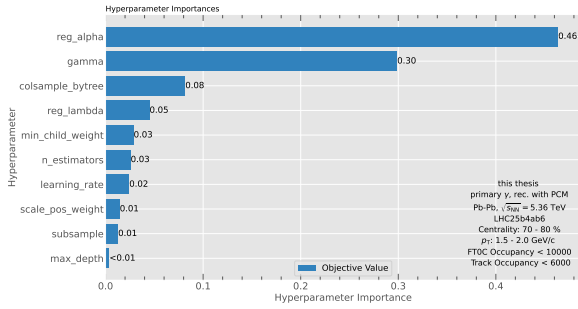
(b) 0.6-0.8 GeV/c



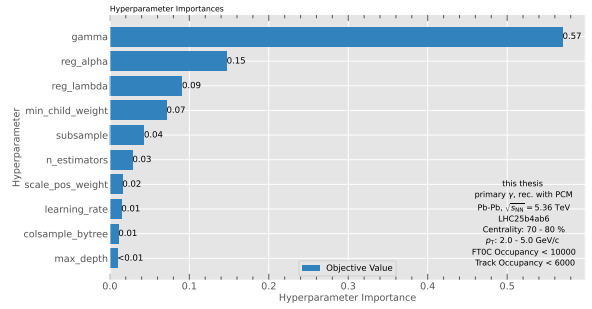
(c) 0.8-1.0 GeV/c



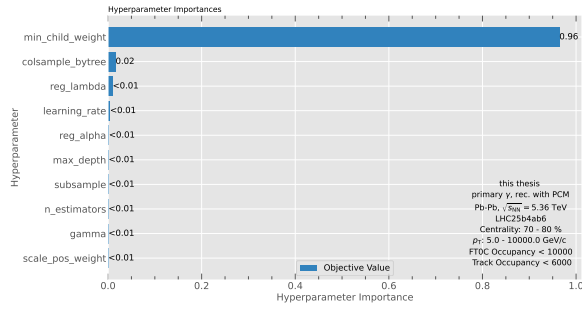
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

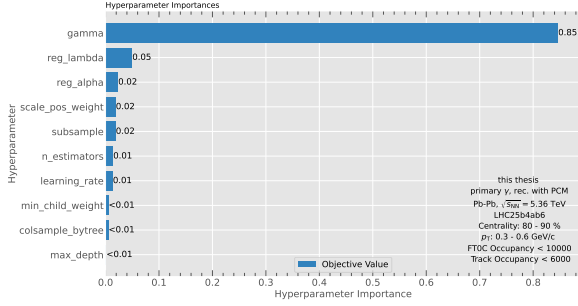


(f) 2.0-5.0 GeV/c

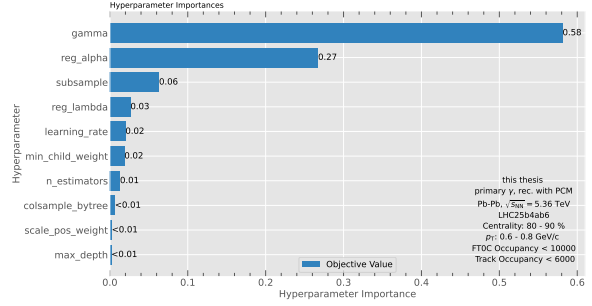


(g) 5.0-10000.0 GeV/c

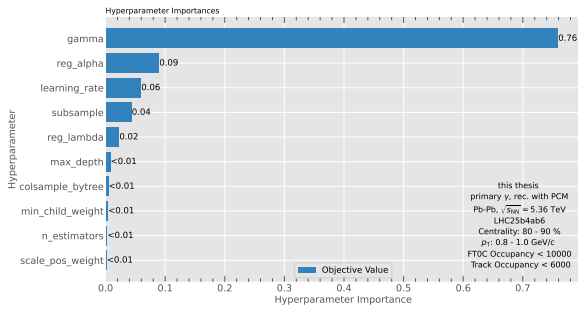
Figure 38: Hyperparameter importance for all p_T bins in the centrality range 70-80 %



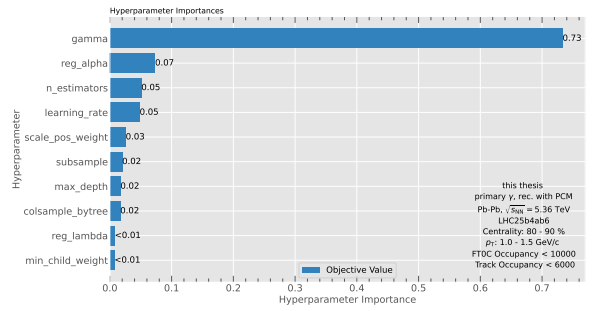
(a) 0.3-0.6 GeV/c



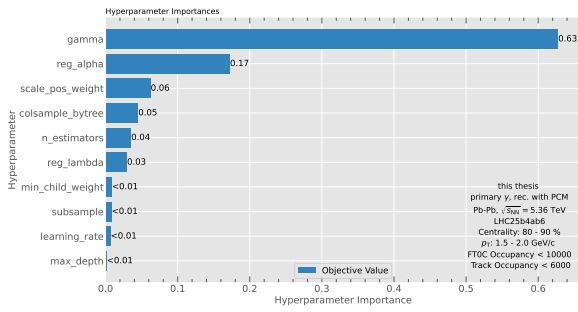
(b) 0.6-0.8 GeV/c



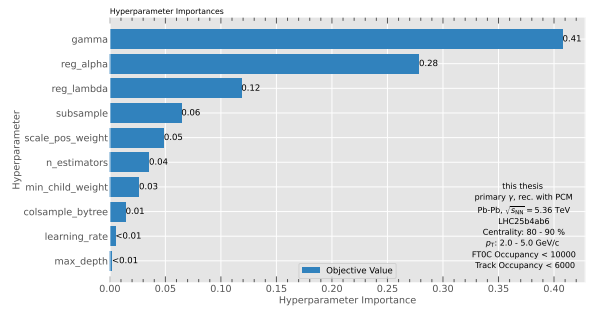
(c) 0.8-1.0 GeV/c



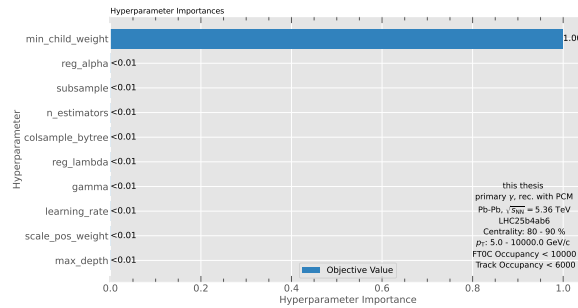
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c

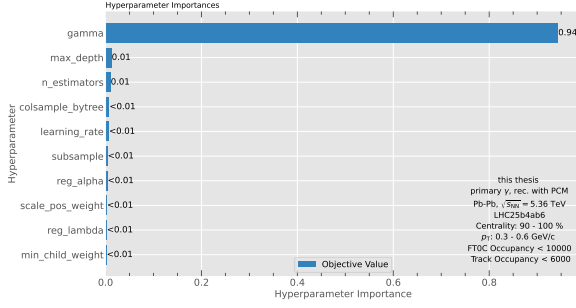


(f) 2.0-5.0 GeV/c

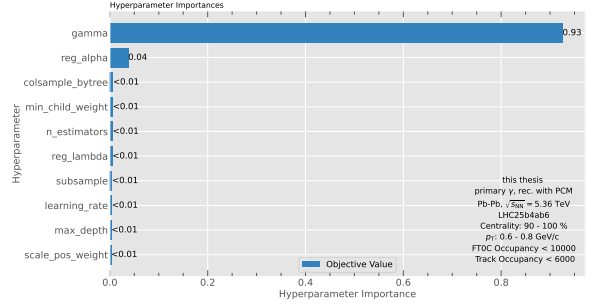


(g) 5.0-10000.0 GeV/c

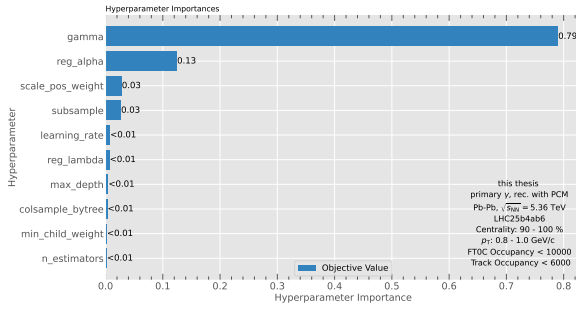
Figure 39: Hyperparameter importance for all p_T bins in the centrality range 80-90 %



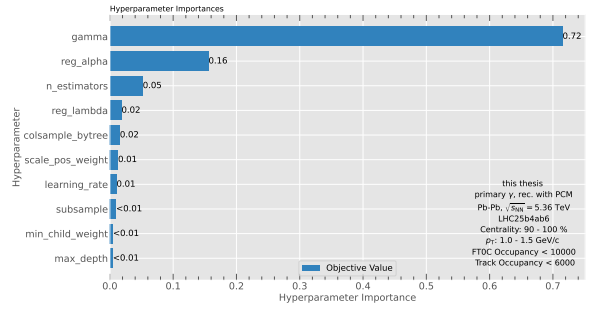
(a) 0.3-0.6 GeV/c



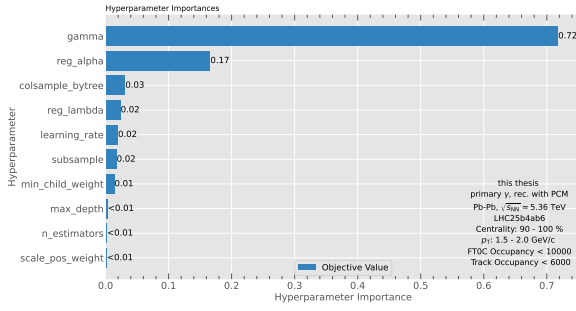
(b) 0.6-0.8 GeV/c



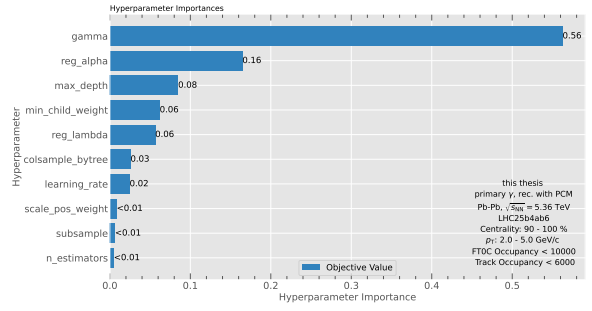
(c) 0.8-1.0 GeV/c



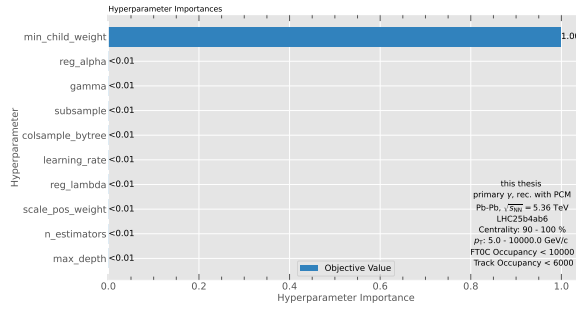
(d) 1.0-1.5 GeV/c



(e) 1.5-2.0 GeV/c



(f) 2.0-5.0 GeV/c



(g) 5.0-10000.0 GeV/c

Figure 40: Hyperparameter importance for all p_T bins in the centrality range 90-100 %